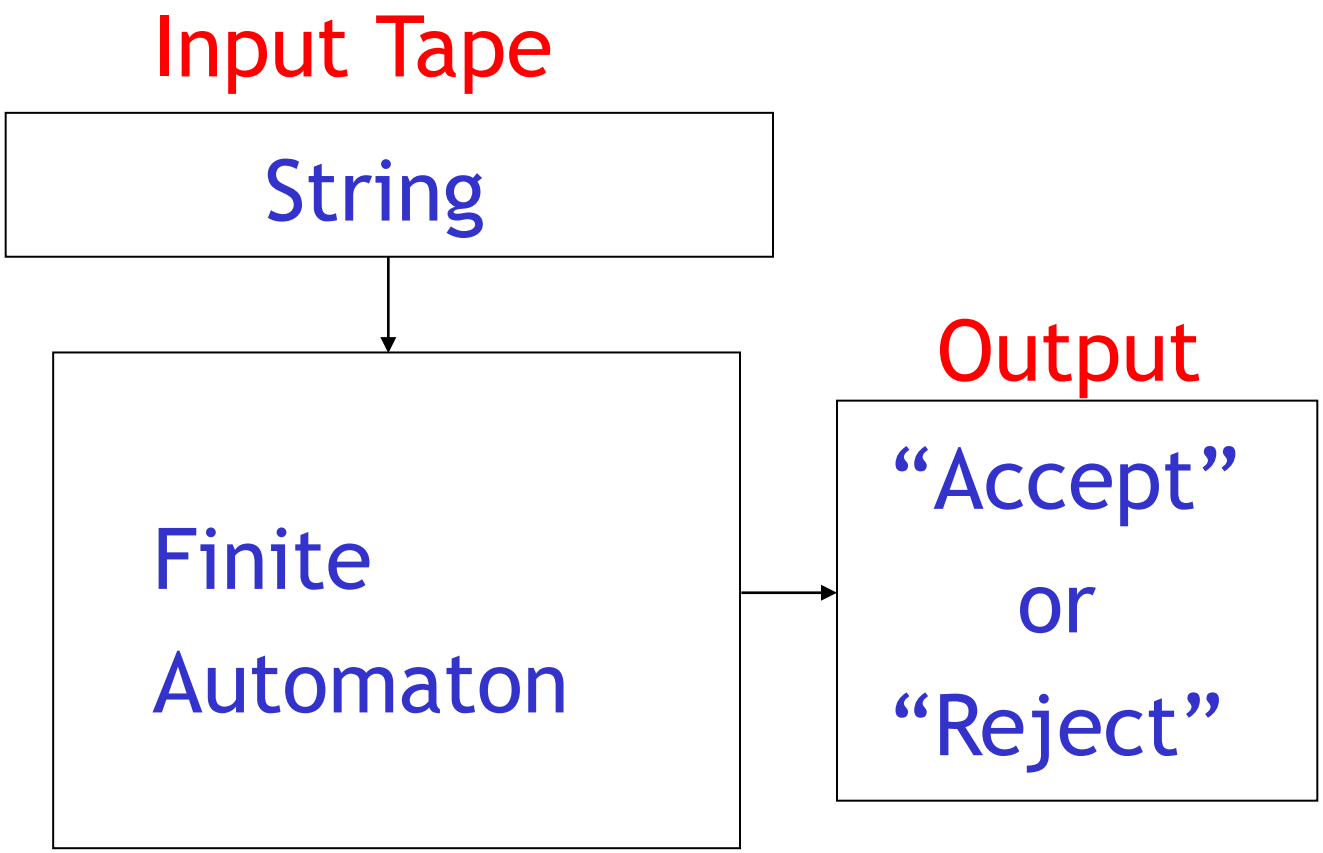
Deterministic

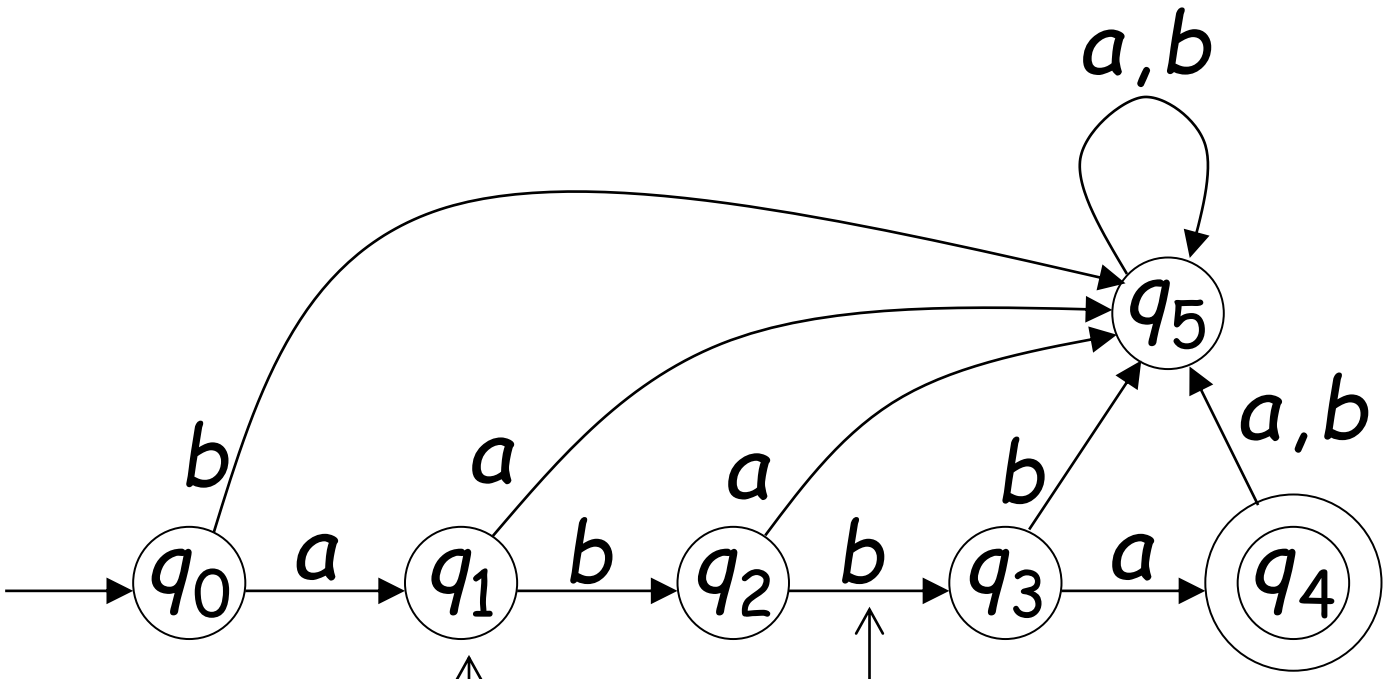
# Finite Automata

And Regular Languages

# Deterministic Finite Automaton (DFA)



# Transition Graph



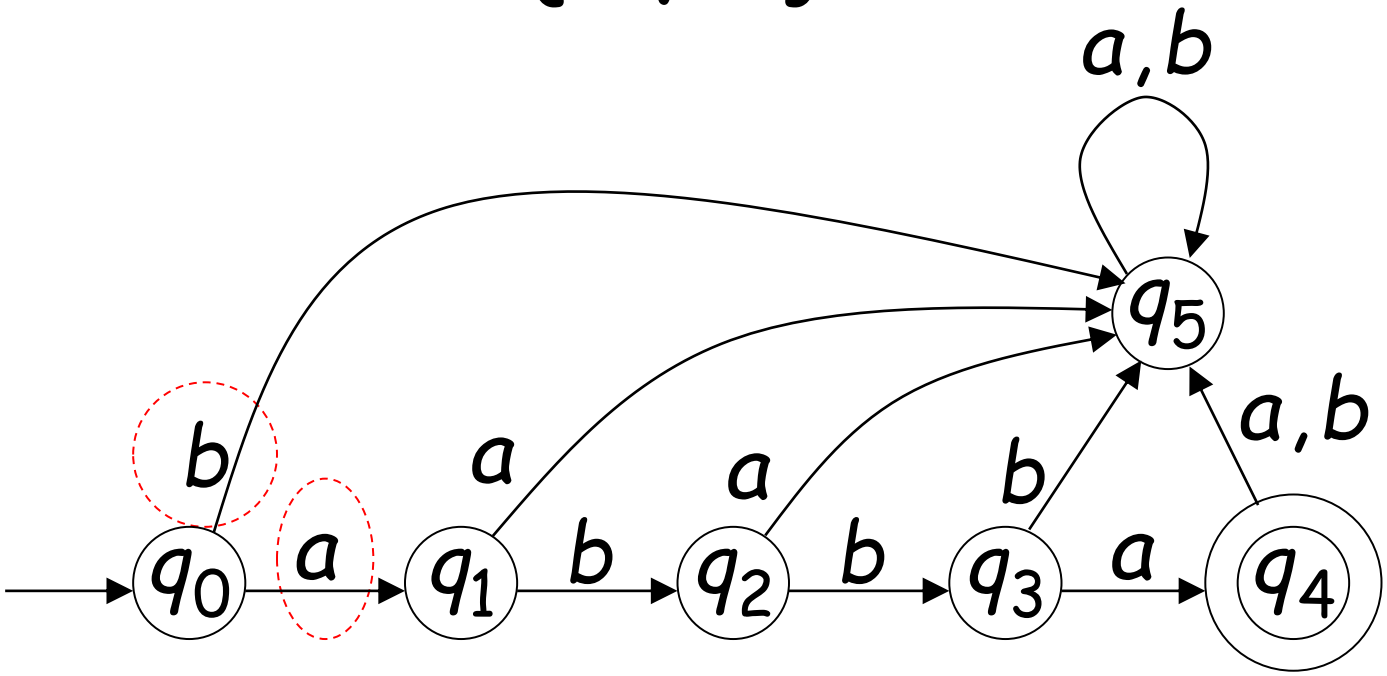

initial state

state

transition

accepting state

Alphabet $\Sigma = \{a, b\}$




For <u>every</u> state, there is a transition

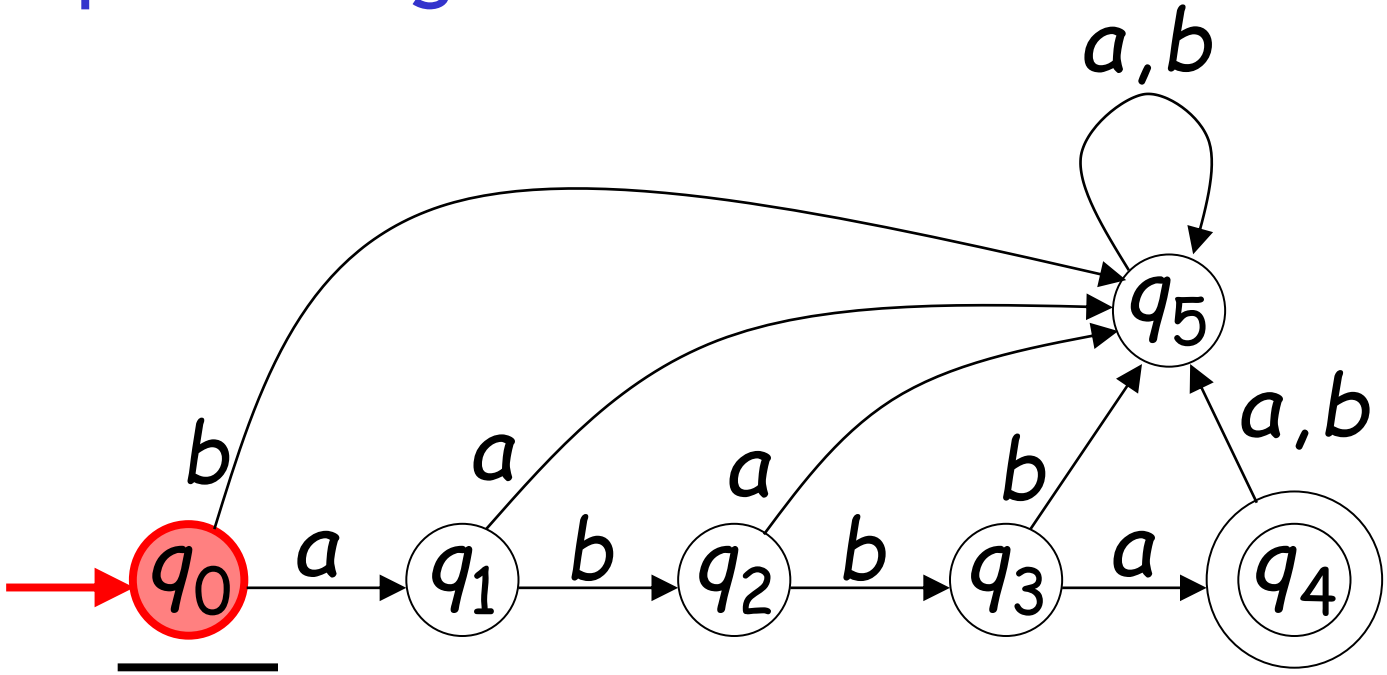
for <u>every</u> symbol in the alphabet

# Initial Configuration

Input Tape

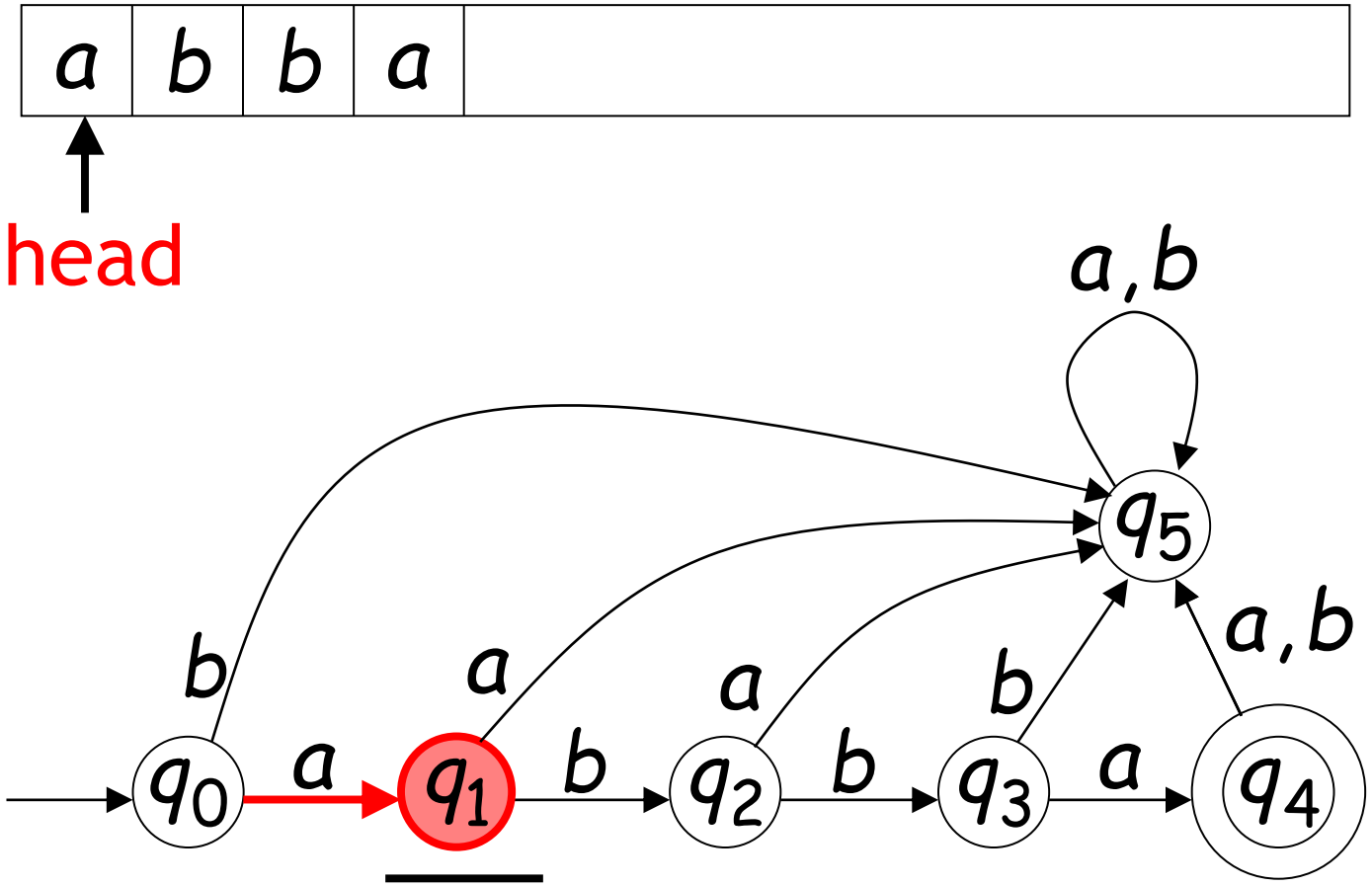
| a | b | b | a | |
|---|---|---|---|---|

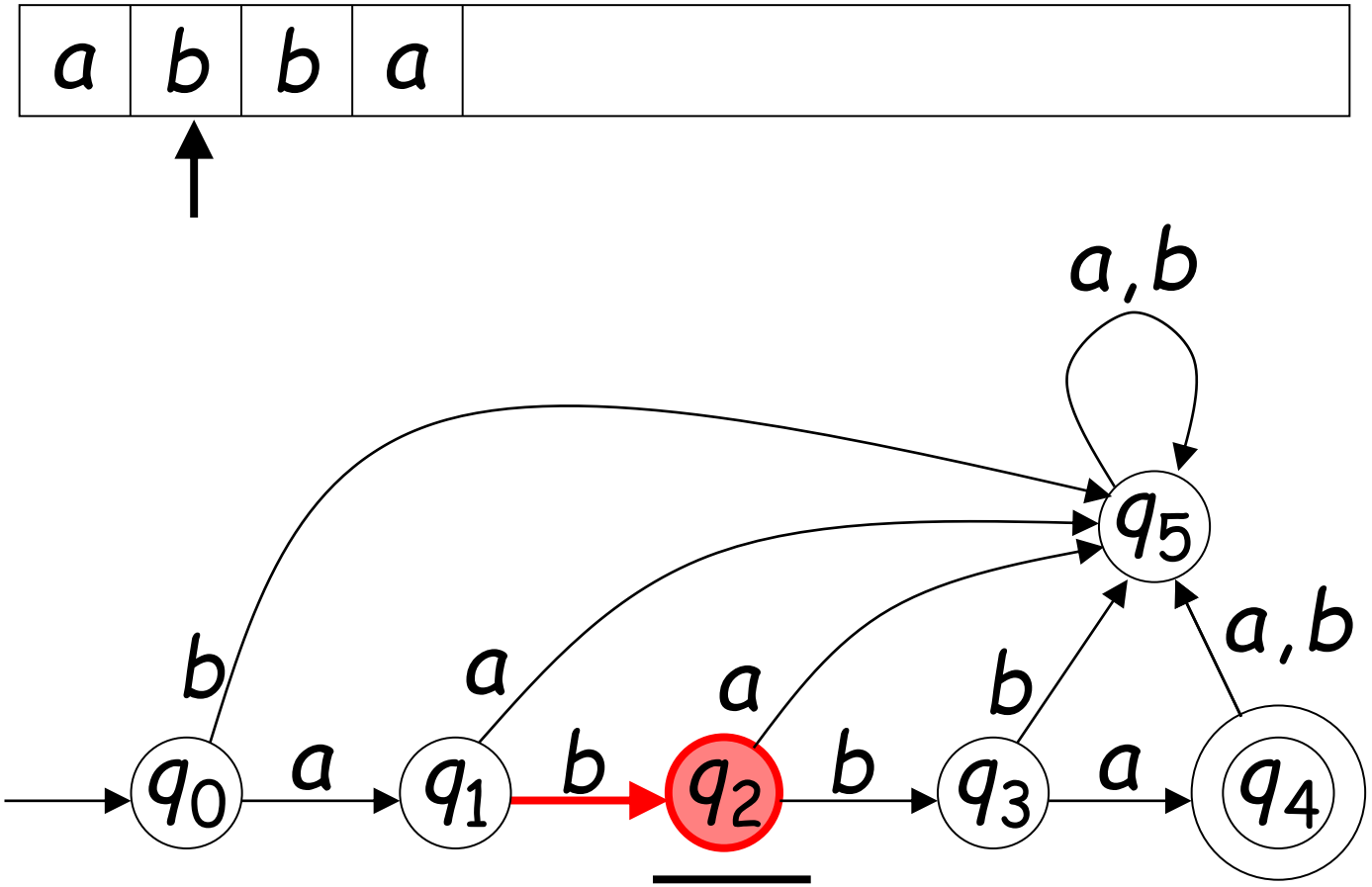
Input String

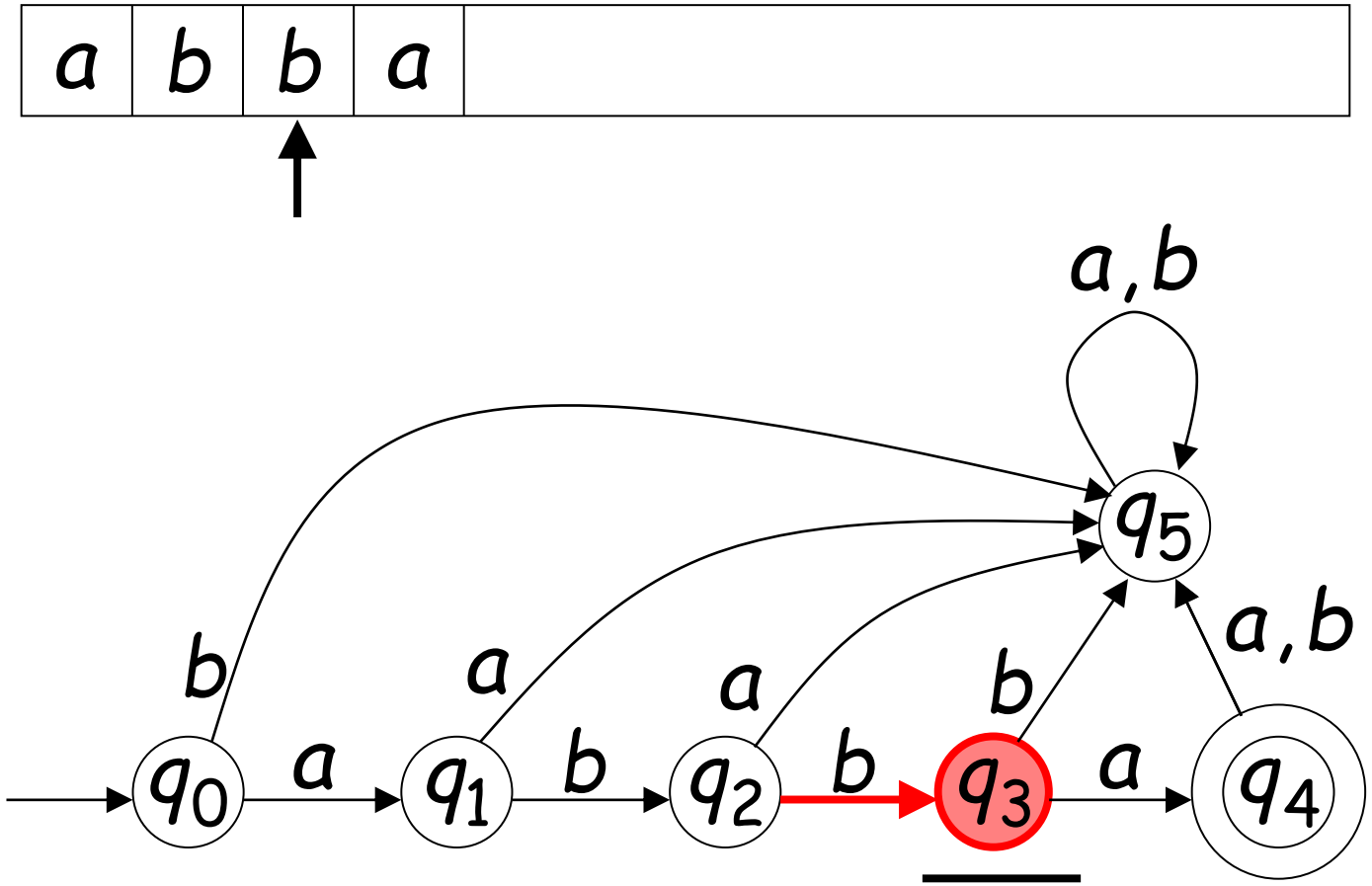



Initial state

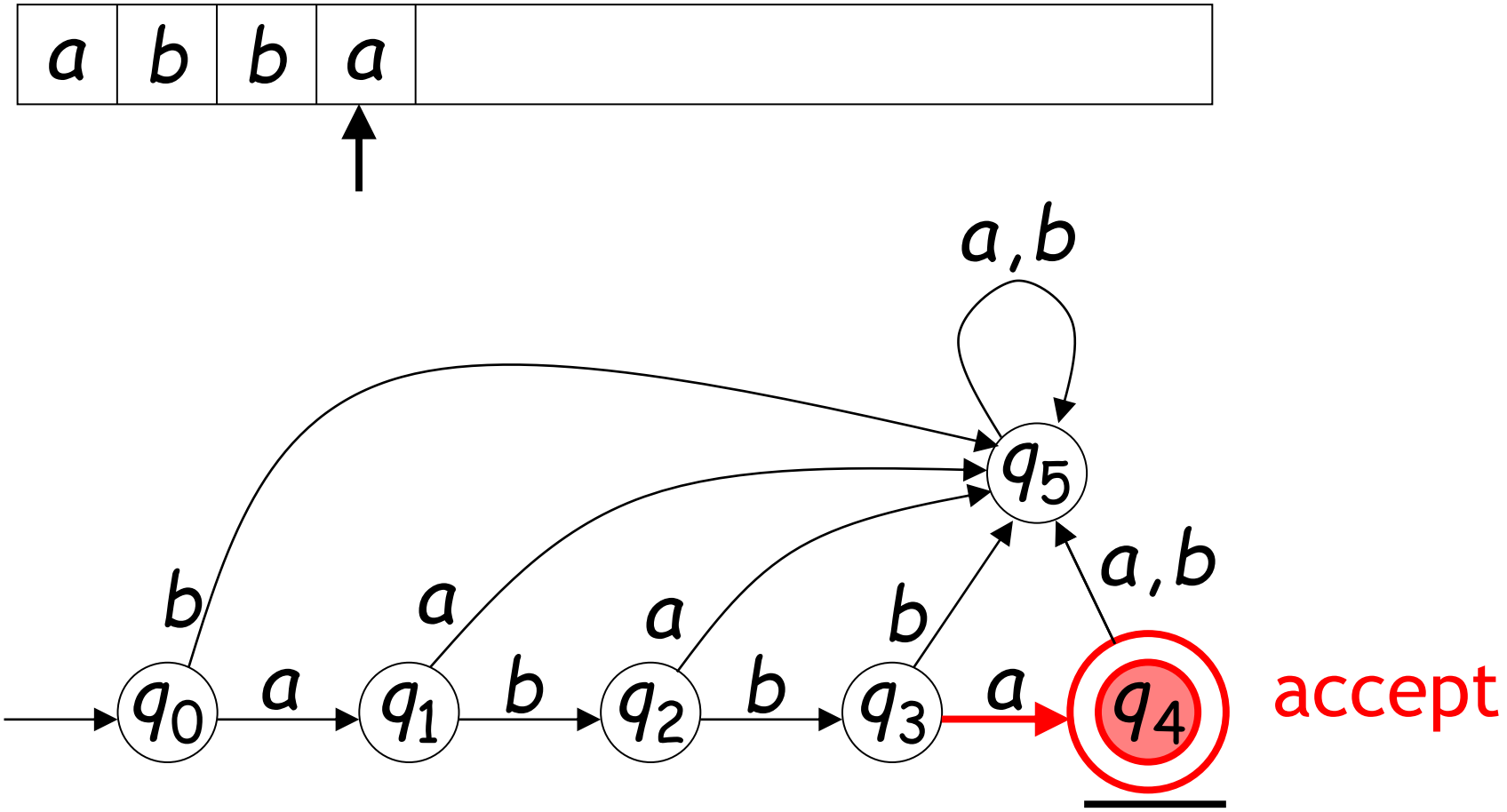
# Scanning the Input




a b b a
a,b
q5
b
a
a
b
a,b
q0
a
q1
b
q2
b
q3
a
q4


a b b a
a,b
q5
b
a
a
b
a,b
q0
a
q1
b
q2
b
q3
a
q4

# Input finished

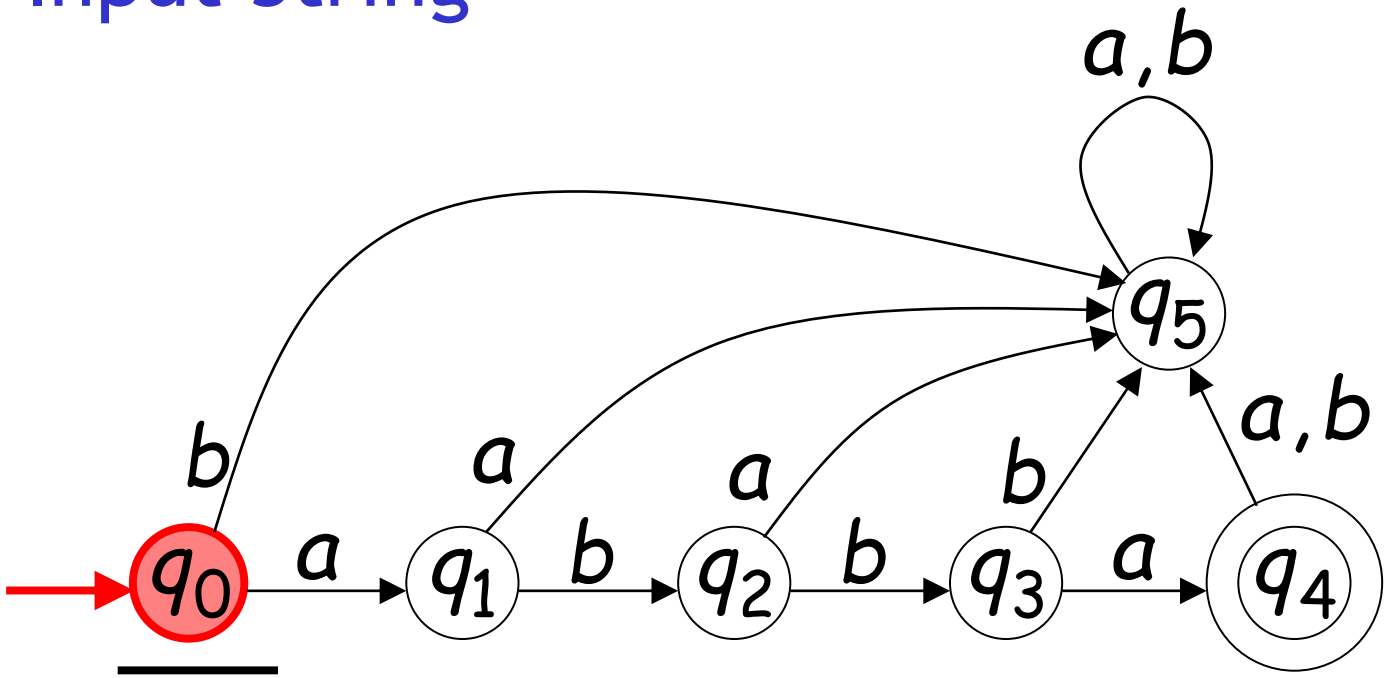


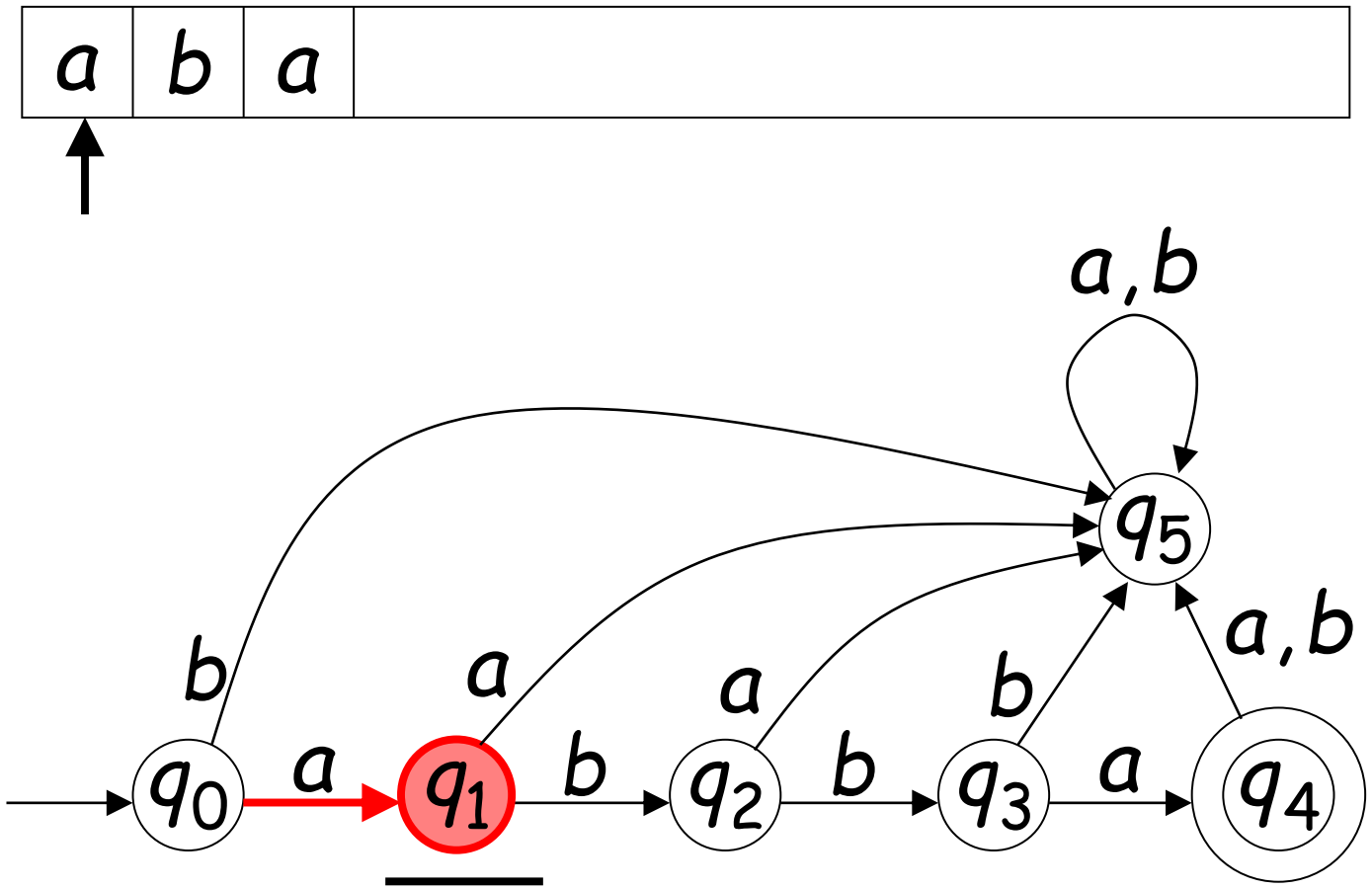

Last state determines the outcome

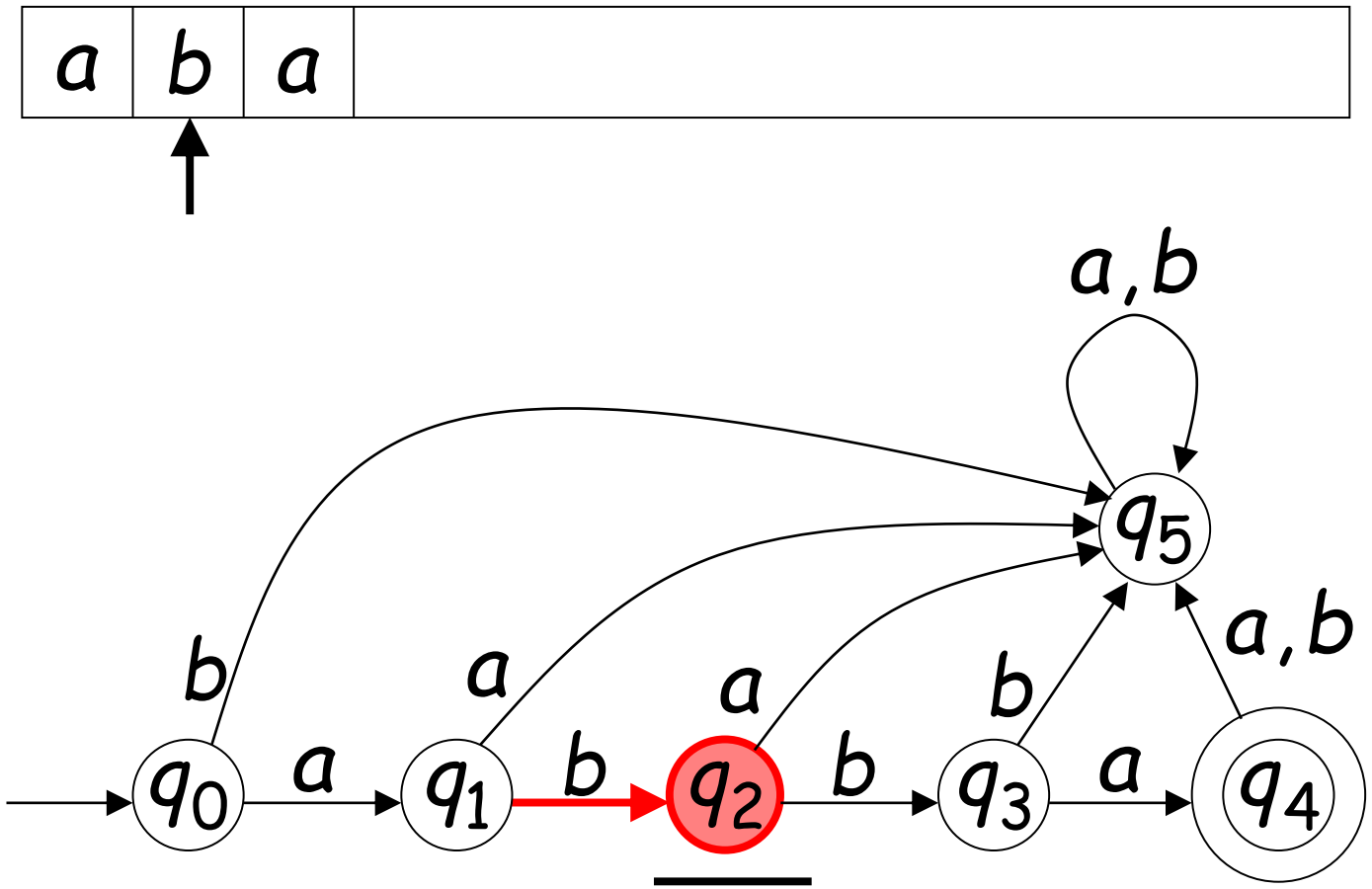
# A Rejection Case

| a | b | a | |
|---|---|---|---|

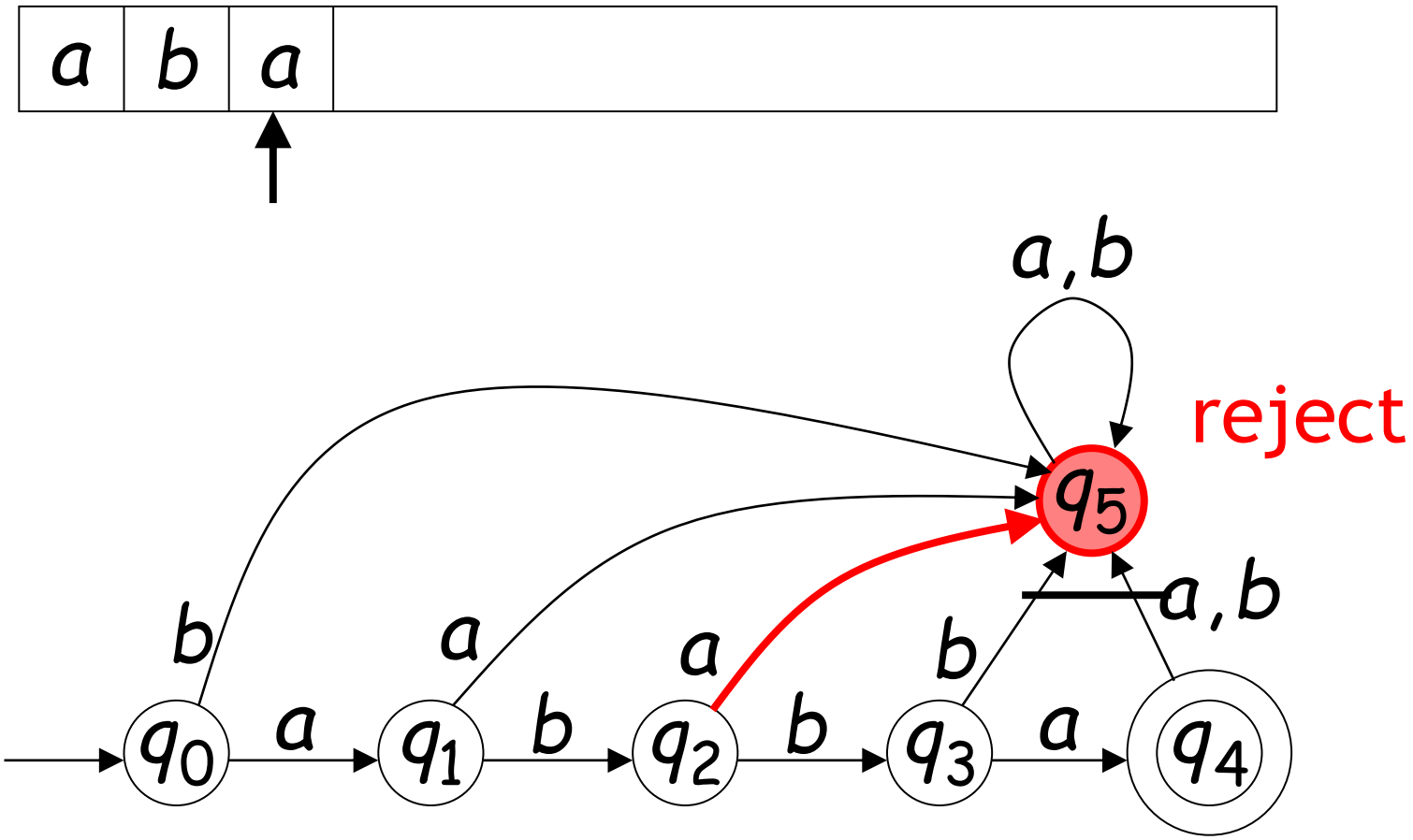
Input String




a b a
a,b
q5
b
a
a
b
a,b
q0
a
q1
b
q2
b
q3
a
q4


a
b
a
a,b
q5
b
a
a
b
a,b
q0
a
q1
b
q2
b
q3
a
q4

# Input finished



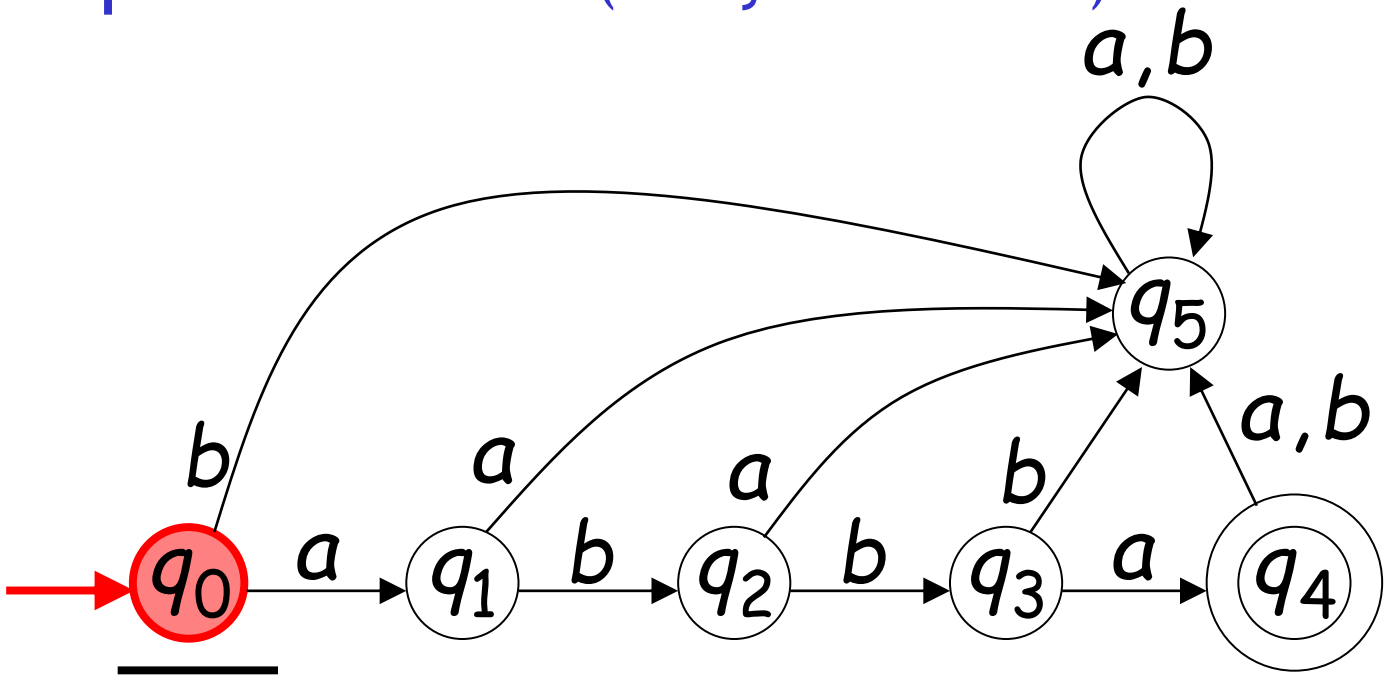

Last state determines the outcome

# Another Rejection Case

Tape is empty

$\varepsilon$

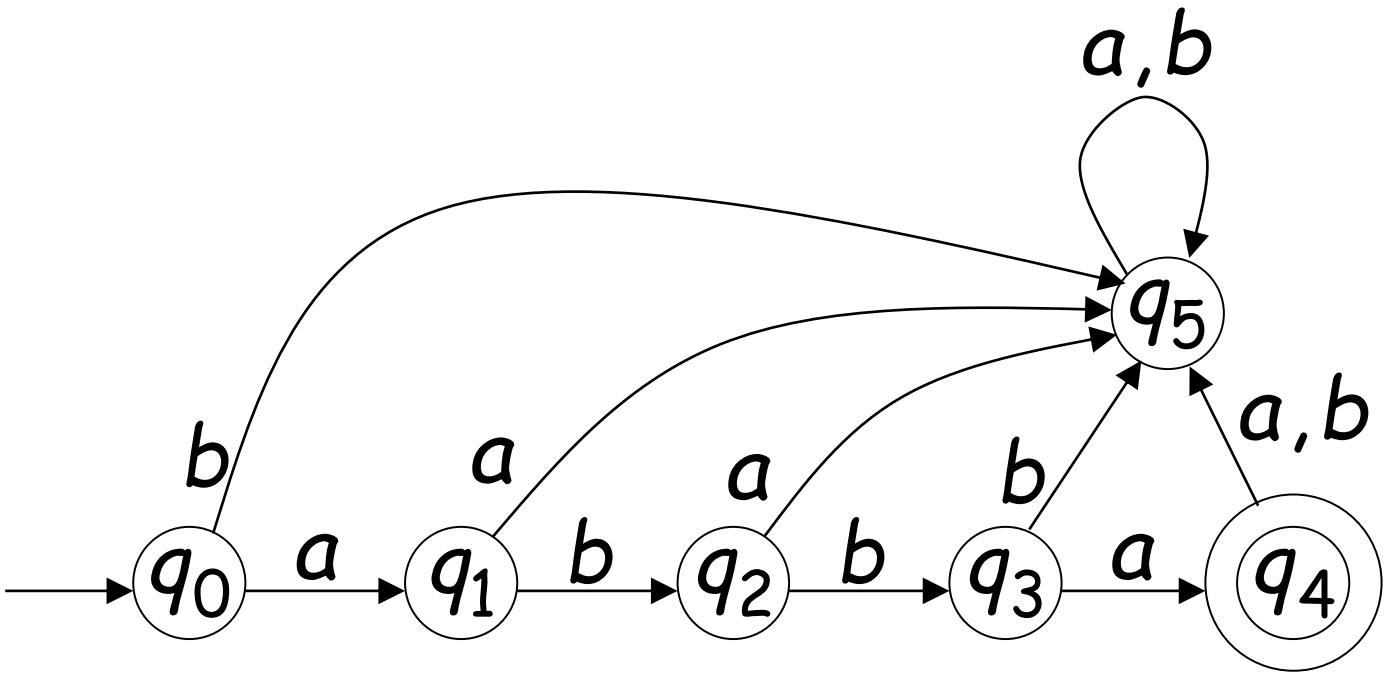
Input Finished (No symbols read)




reject

# This automaton accepts only one string

Language Recognized: $L = \{abba\}$



To accept a string:

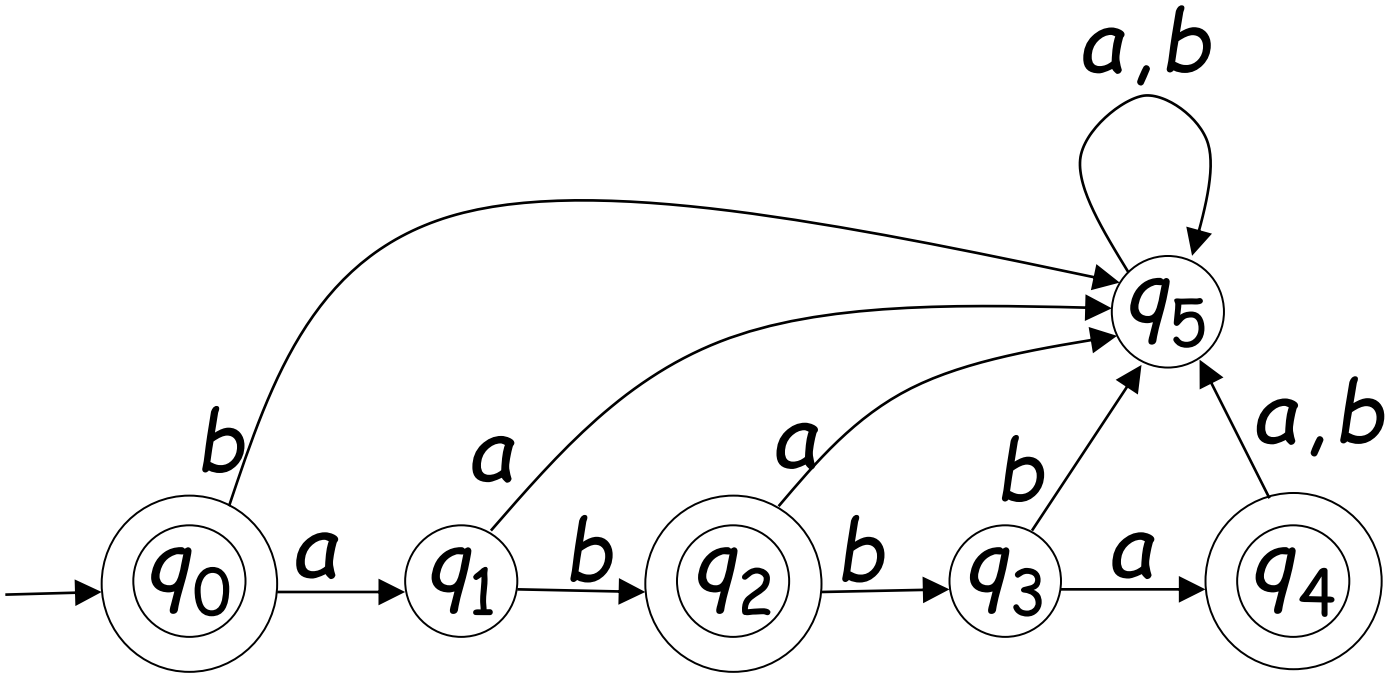
All the input string is scanned
and the last state is an accept state

To reject a string:

All the input string is scanned
and the last state is a non-accept state

# Another Example

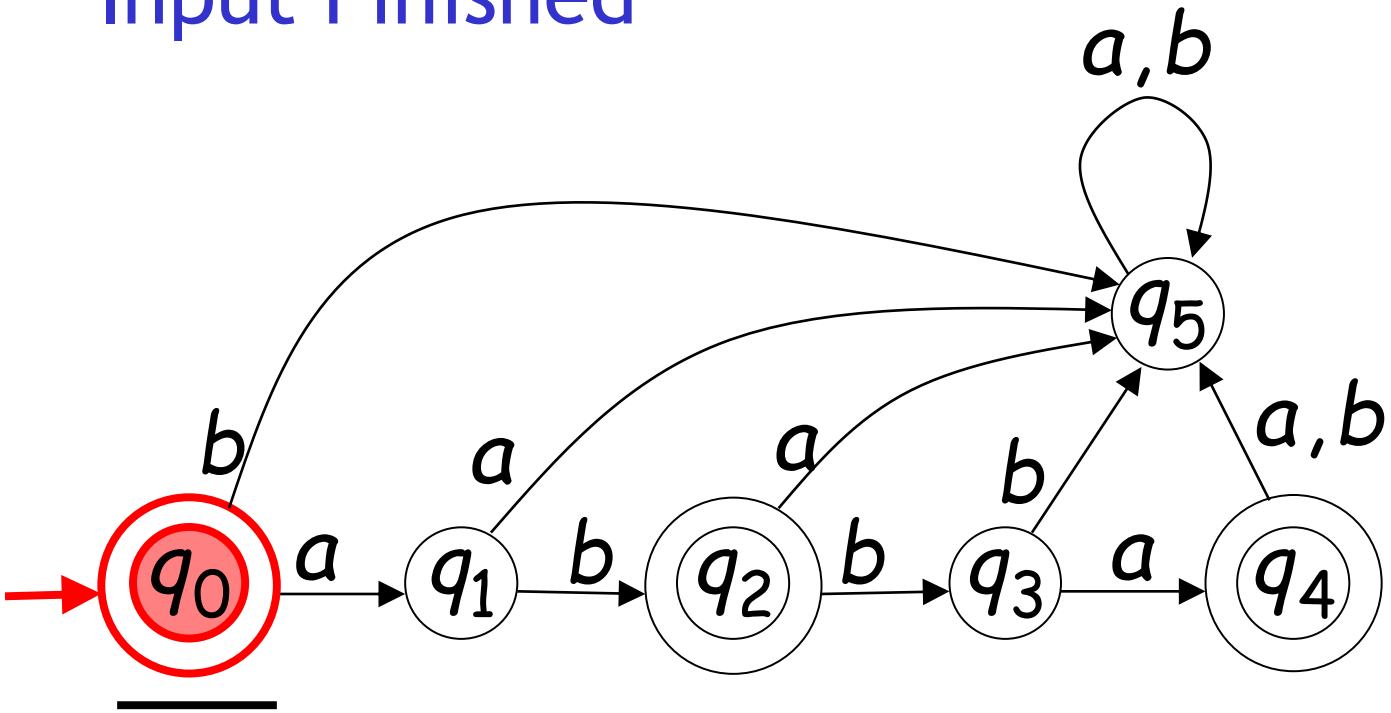
$$L = \{\varepsilon, ab, abba\}$$



# Empty Tape

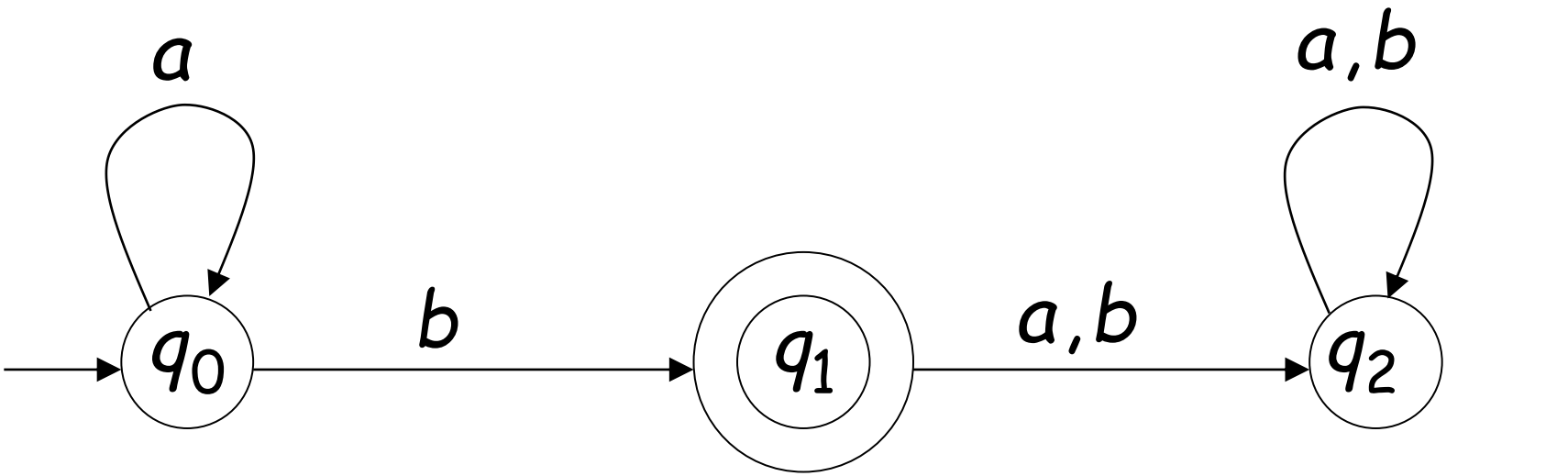
$\varepsilon$

Input Finished



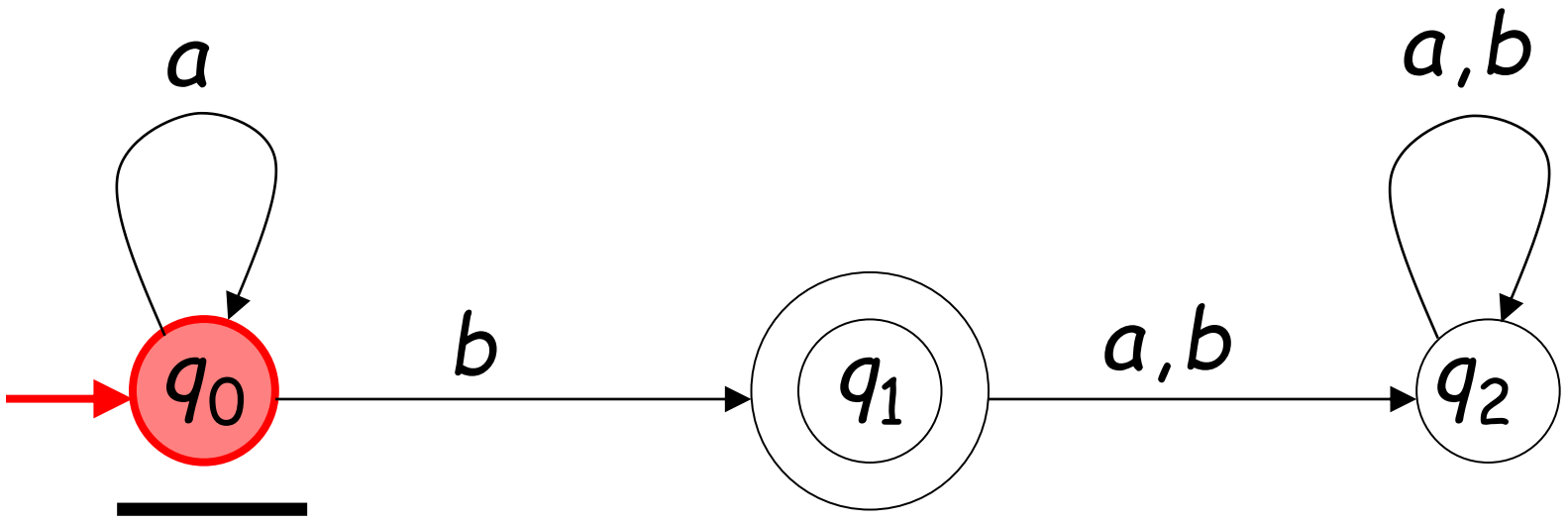

accept

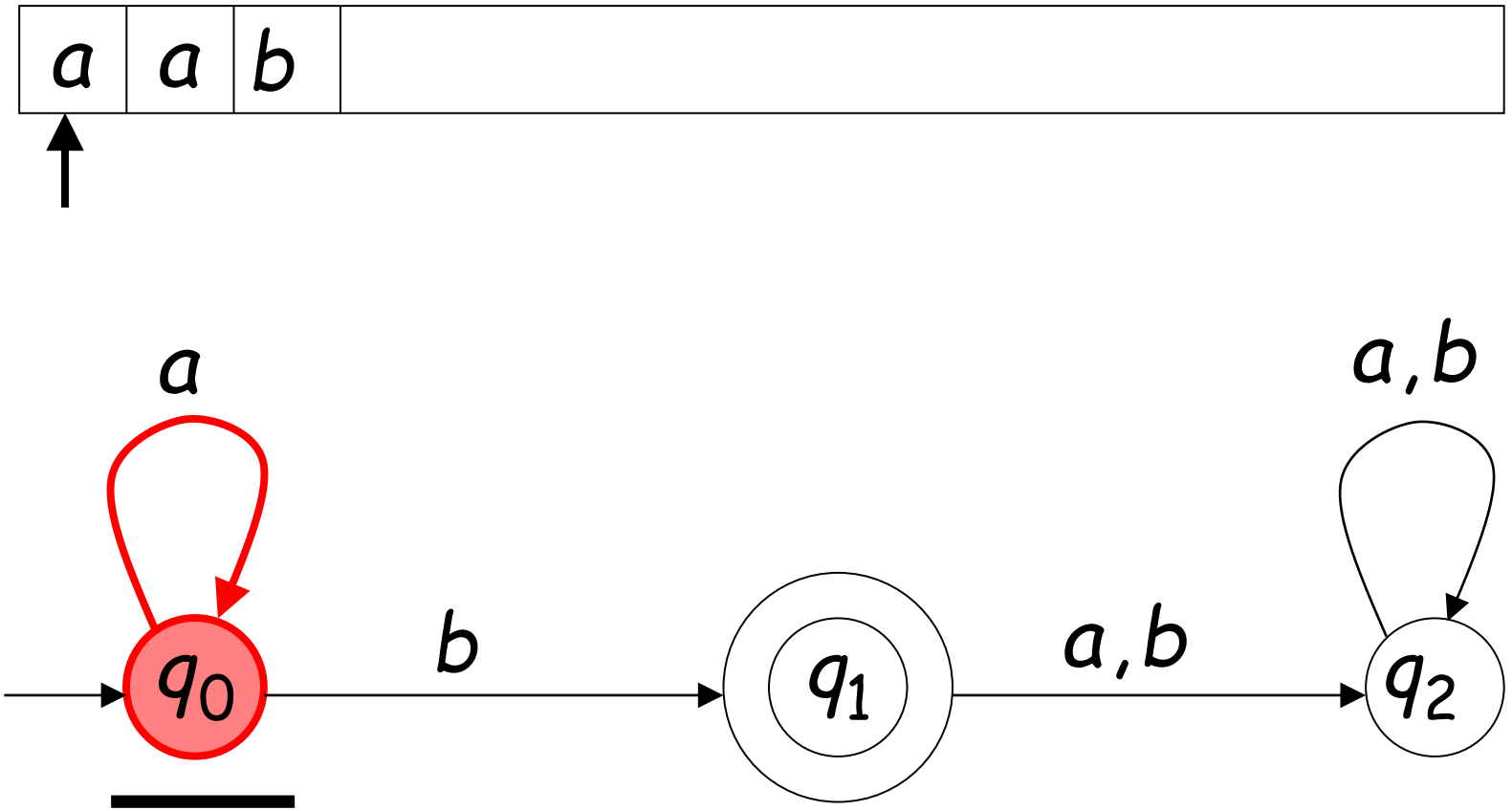
# Another Example

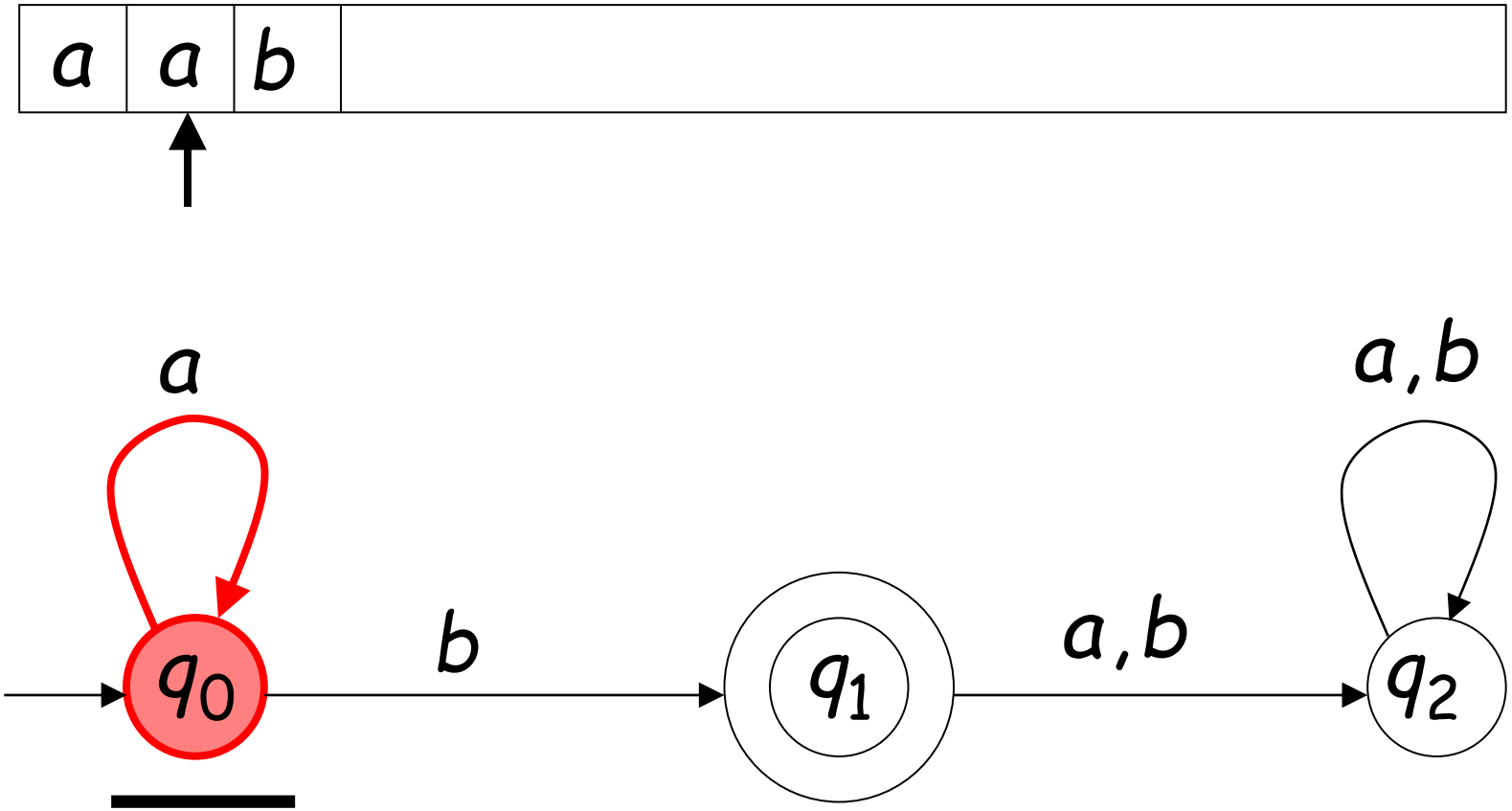


| a | a | b | |
|---|---|---|---|

Input String

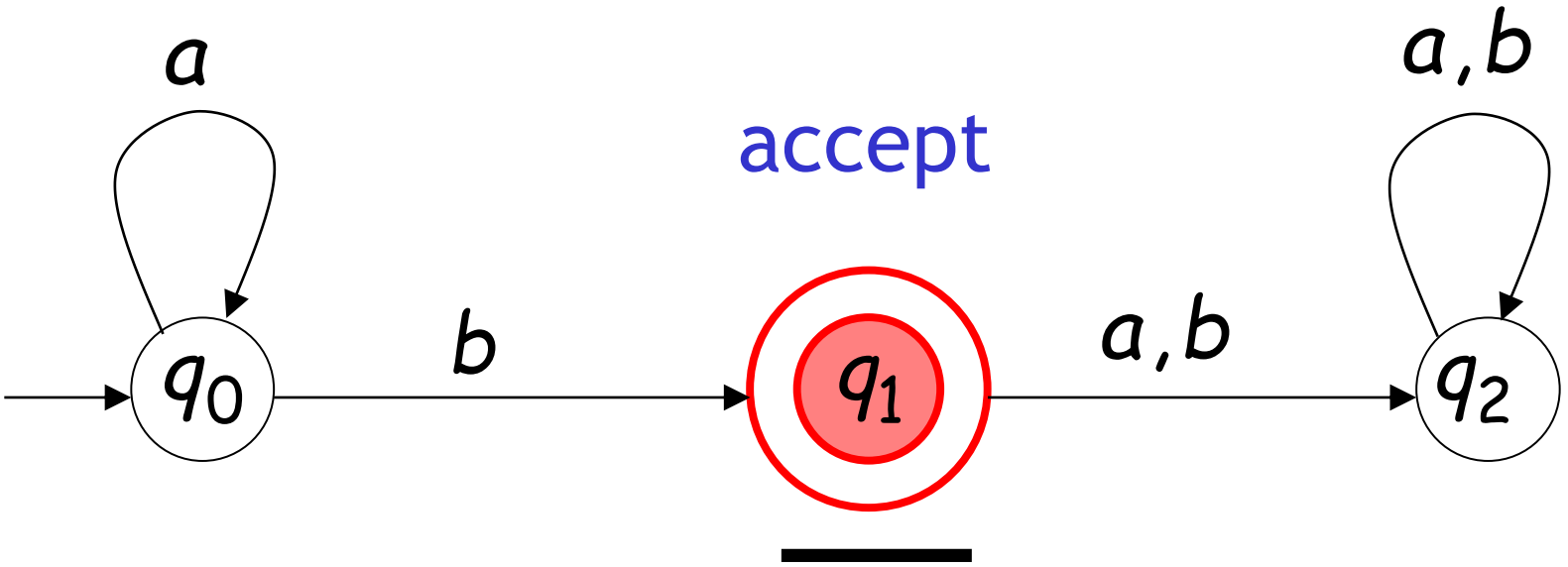



a
a
b
a
a,b
b
a,b
$q_0$
$q_1$
$q_2$


a
a
b
a
a,b
b
a,b
$q_0$
$q_1$
$q_2$

# Input finished

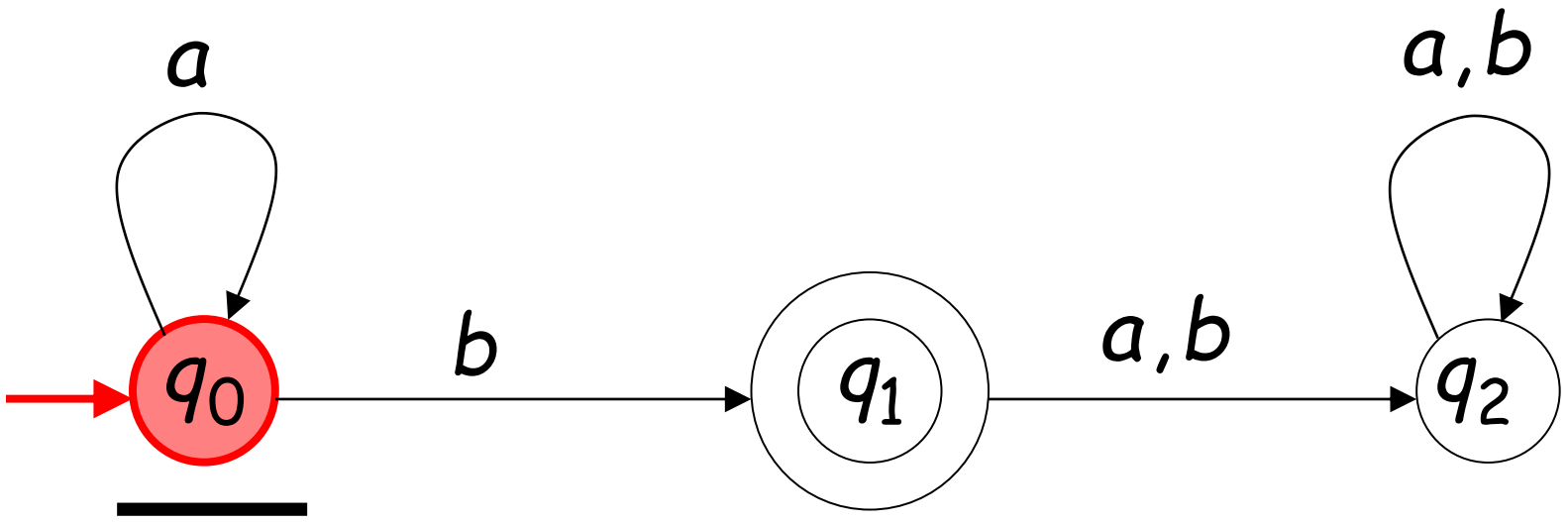


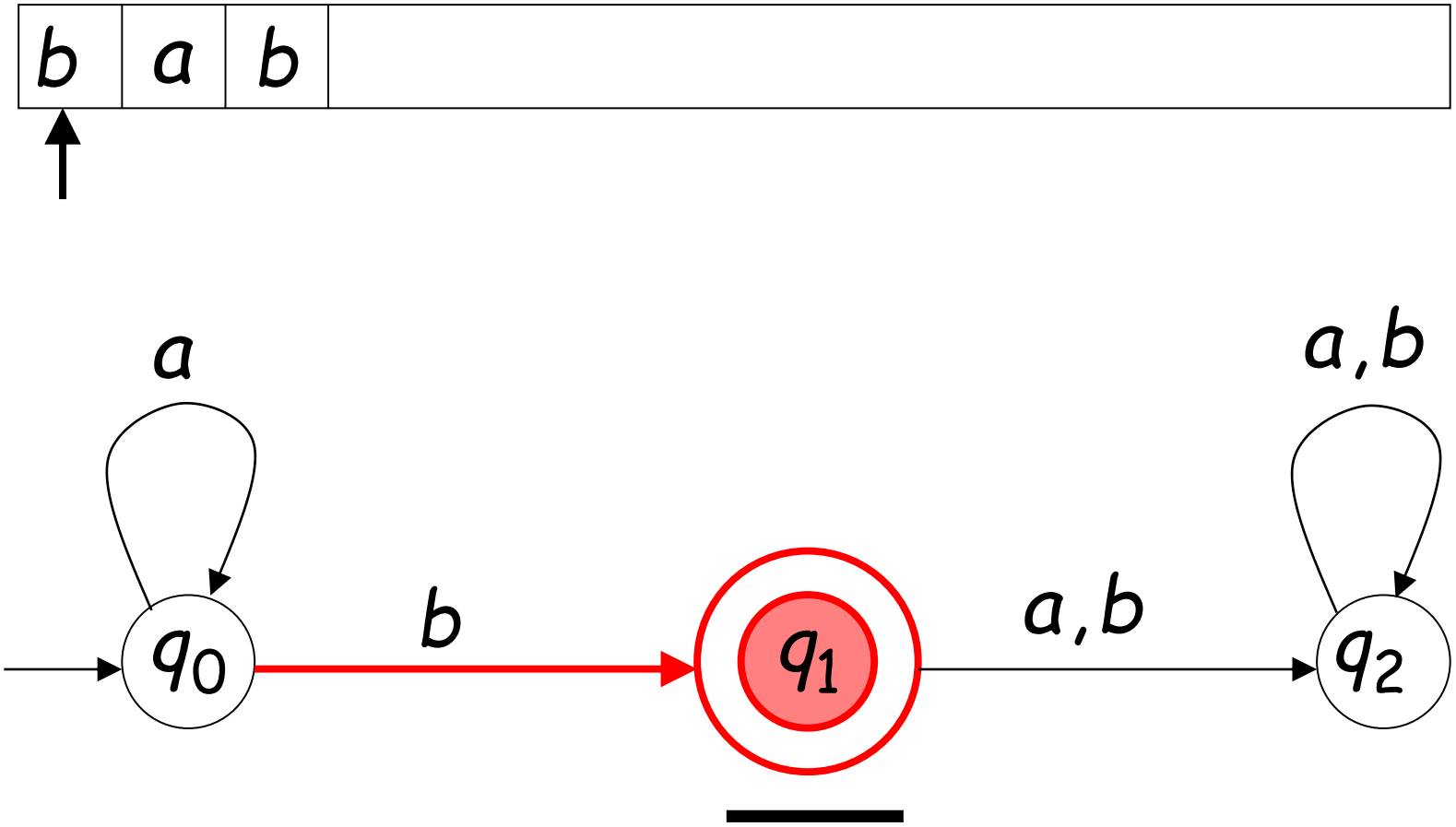



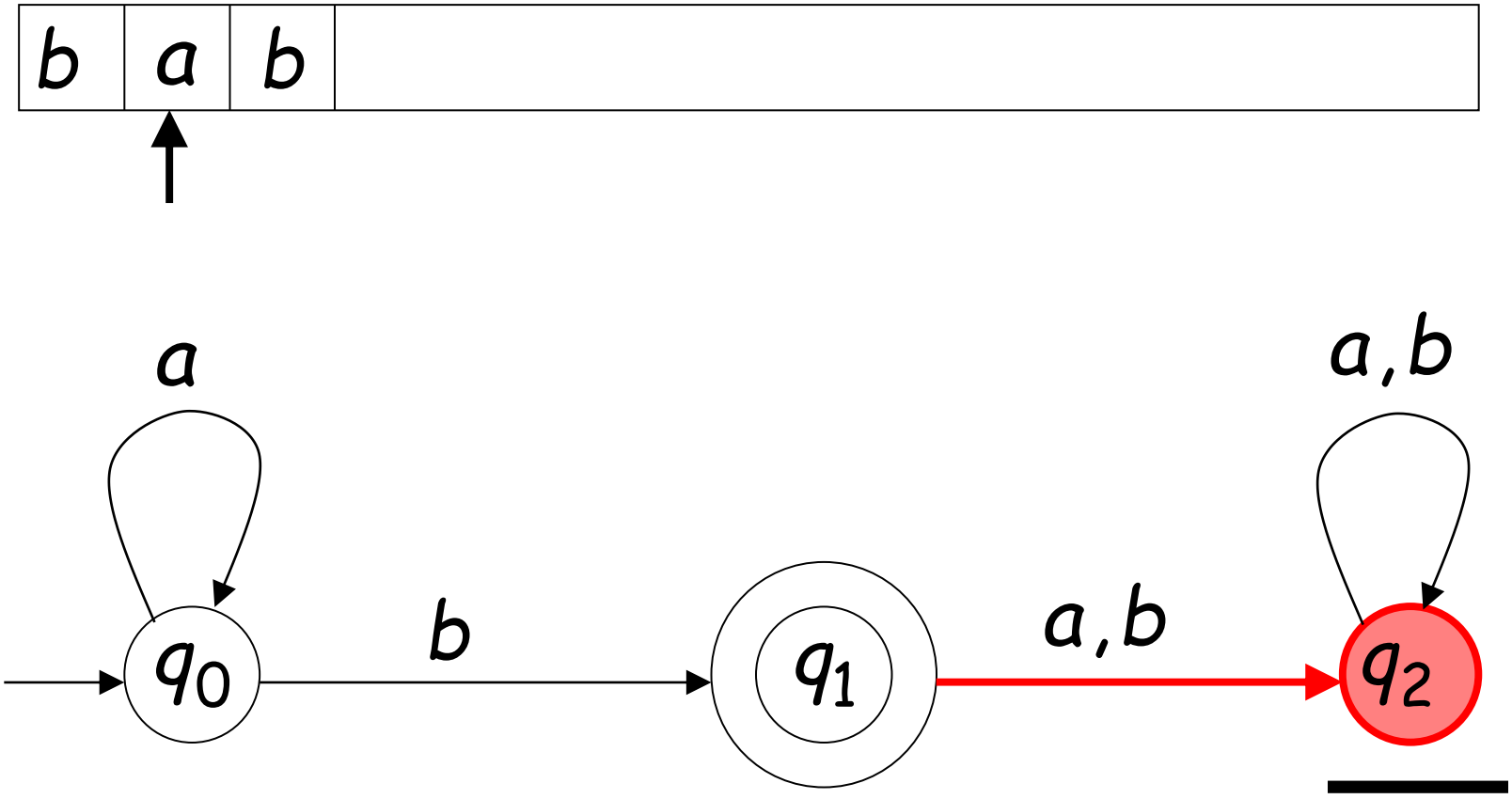
# A rejection case

| b | a | b | |
|---|---|---|---|

Input String

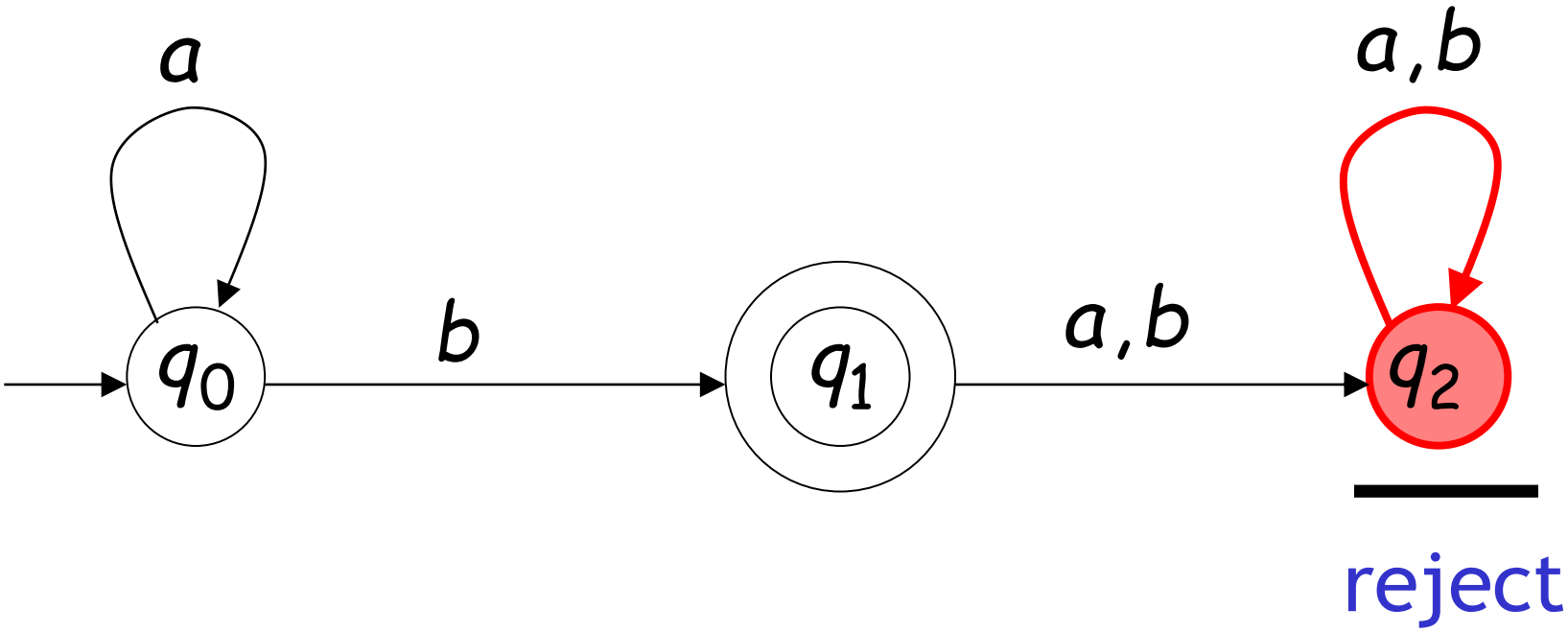



b
a
b
a
a,b
q0
b
q1
a,b
q2


b
a
b
a
a,b
q0
b
q1
a,b
q2

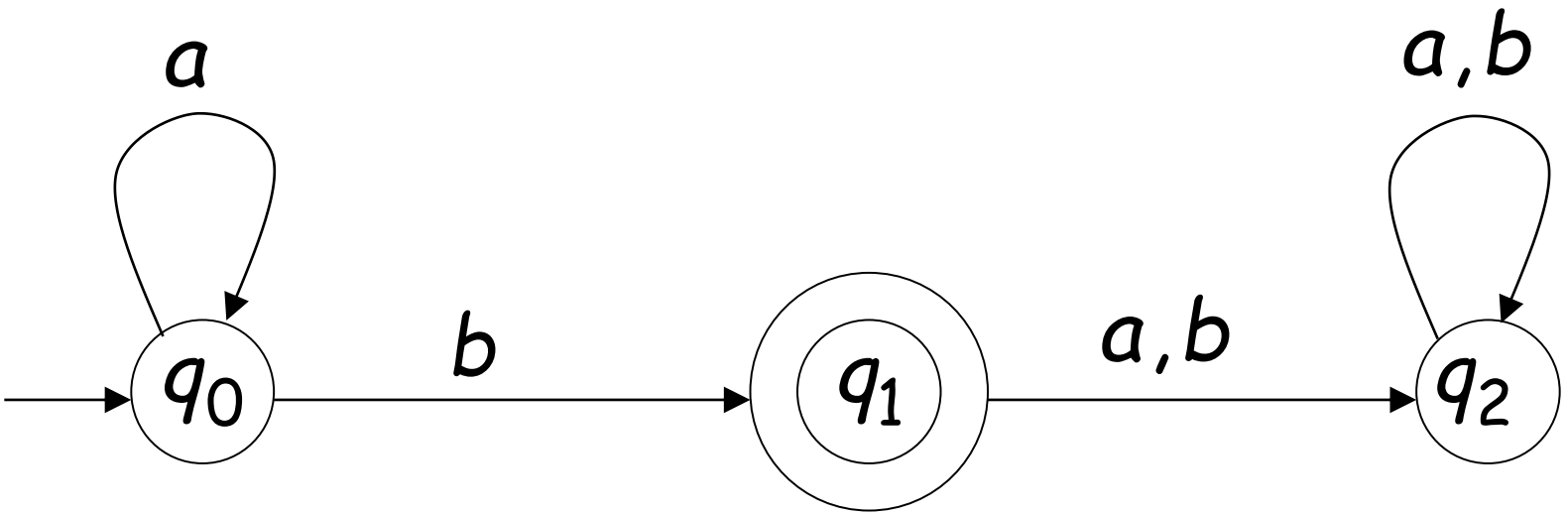
# Input finished



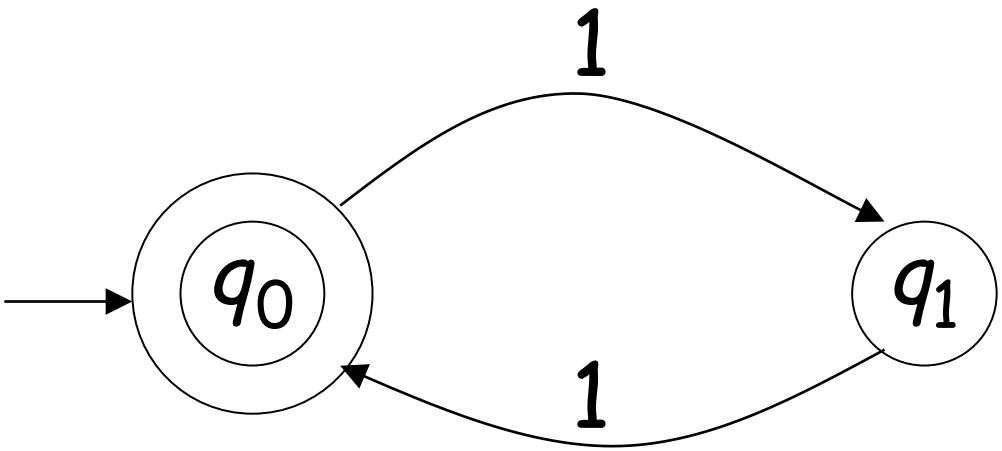

reject

# Language Recognized: $L = \{a^n b : n \geq 0\}$



# Another Example

Alphabet: $\Sigma = \{1\}$




Language Recognized:

$$EVEN = \{x : x \in \Sigma^* \text{ and } x \text{ is even}\}$$

$$= \{\varepsilon, 11, 1111, 111111, \ldots\}$$

# Formal Definition

## Deterministic Finite Automaton (DFA)

$$M = (Q, \Sigma, \delta, q_0, F)$$

$Q$ : set of states

$\Sigma$ : input alphabet $\quad \varepsilon \notin \Sigma$

$\delta$ : transition function

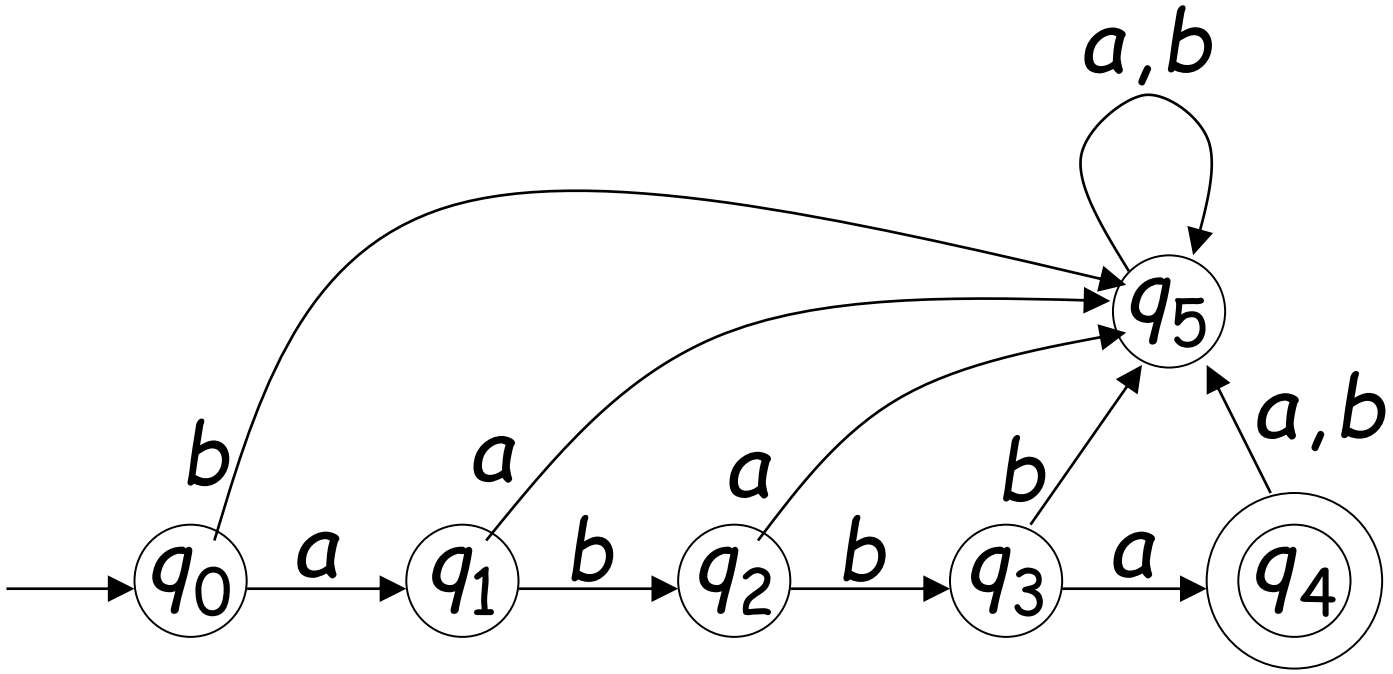
$q_0$ : initial state

$F$ : set of accepting states

# Set of States $Q$

## Example

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$$

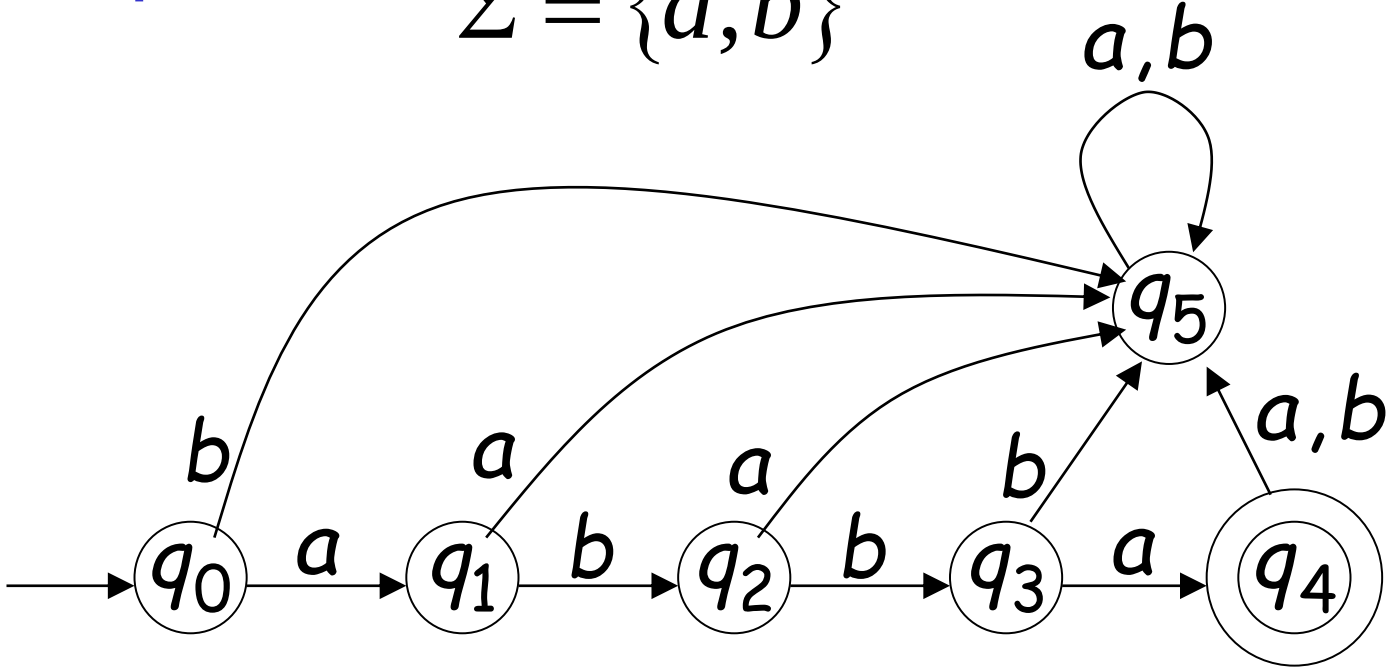


# Input Alphabet $\Sigma$

$\varepsilon \notin \Sigma$ :the input alphabet never contains $\varepsilon$
empty string

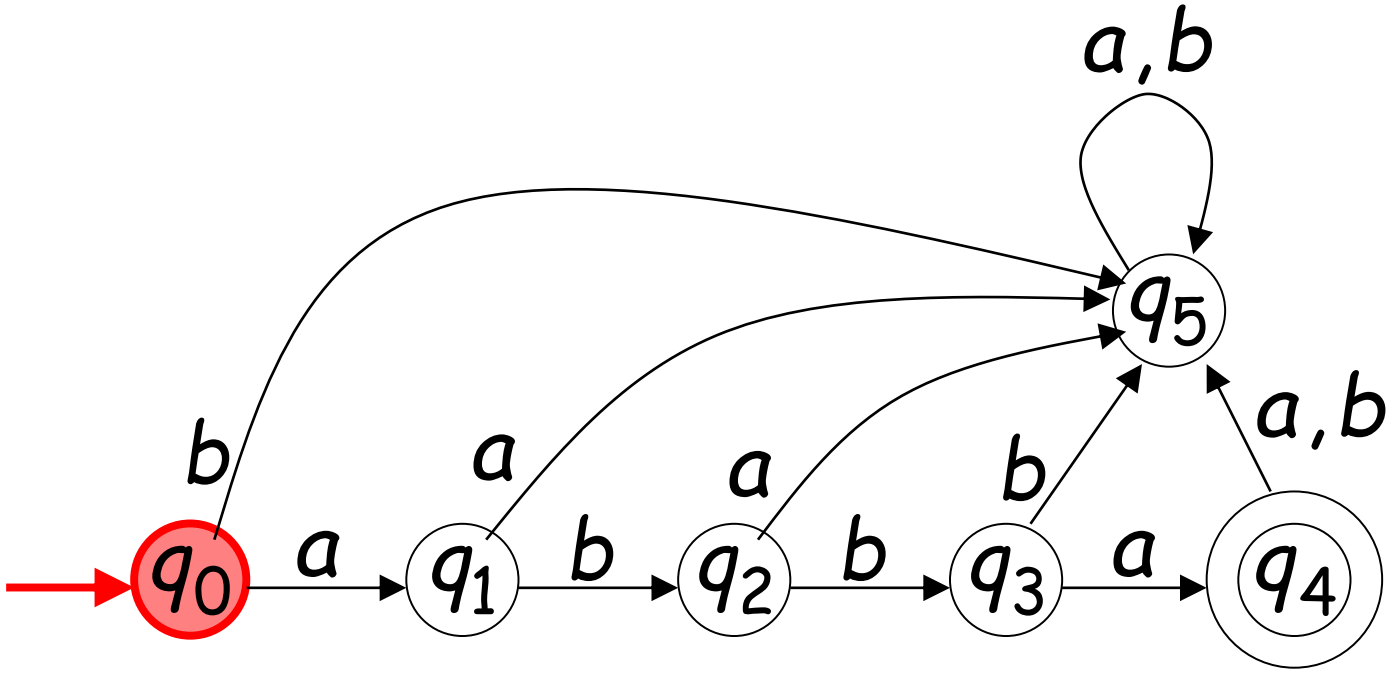
Example

$$\Sigma = \{a, b\}$$



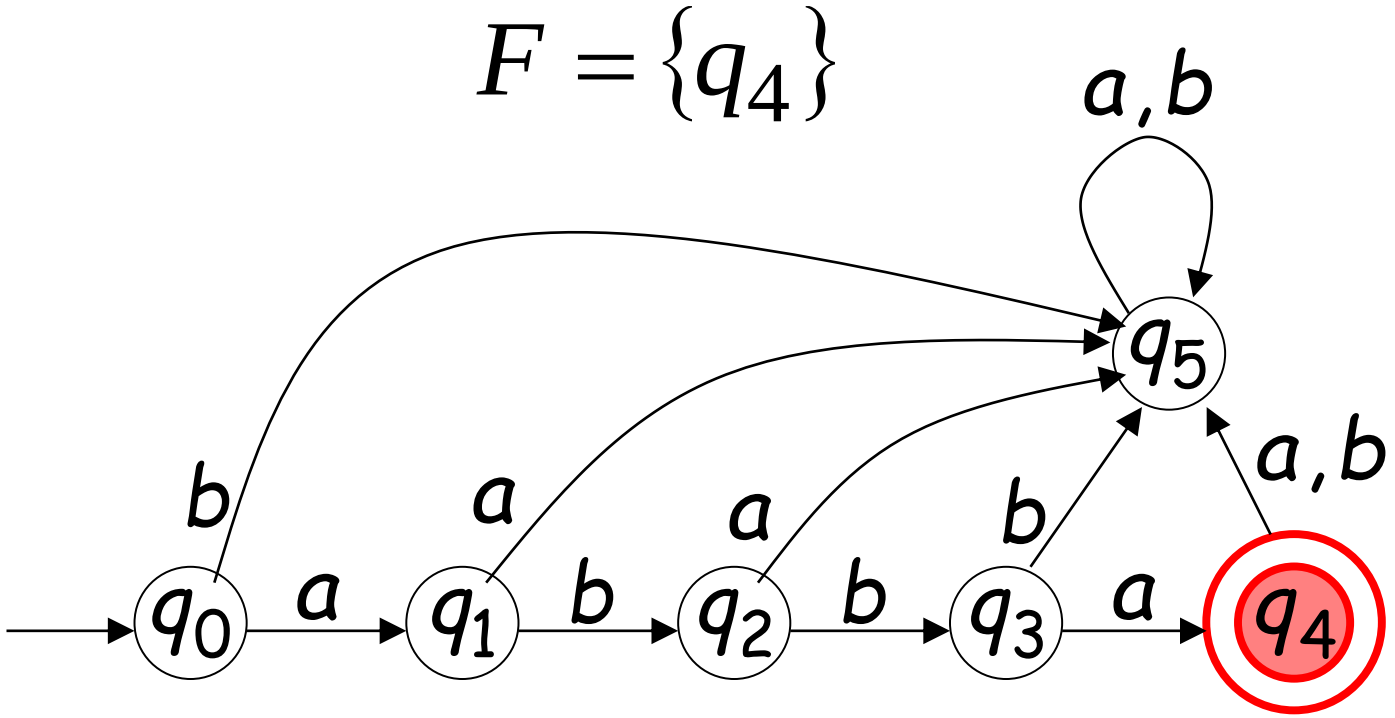
# Initial State $q_0$

## Example



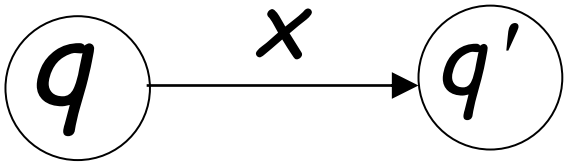
# Set of Accepting States $F \subseteq Q$

## Example



# Transition Function $\delta : Q \times \Sigma \rightarrow Q$

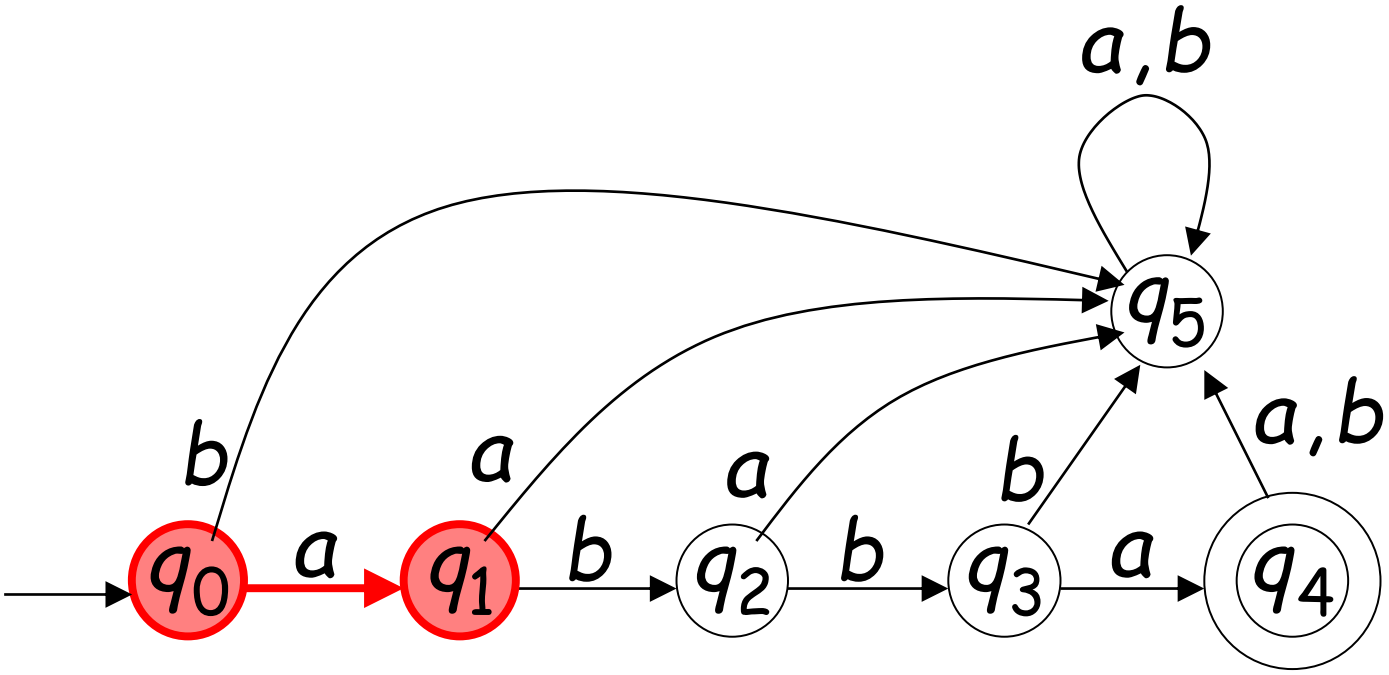
$$\delta(q, x) = q'$$



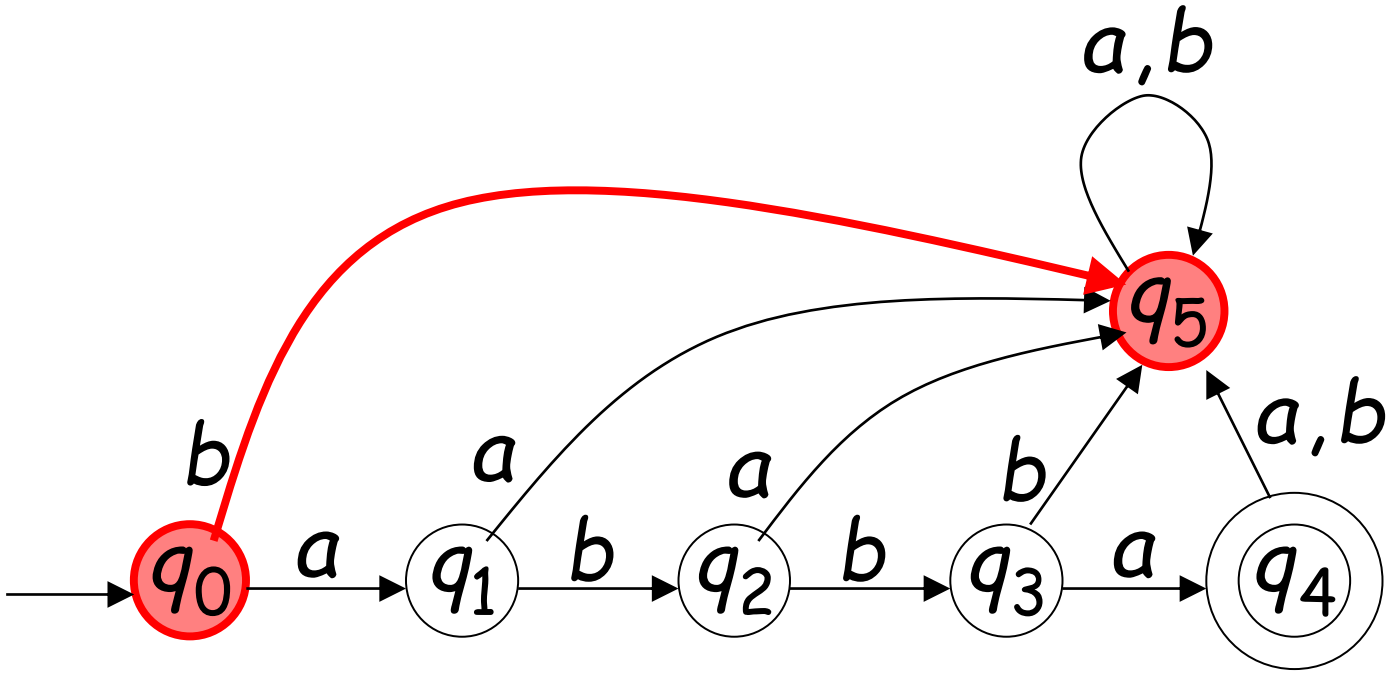

Describes the result of a transition from state $q$ with symbol $x$

Example:

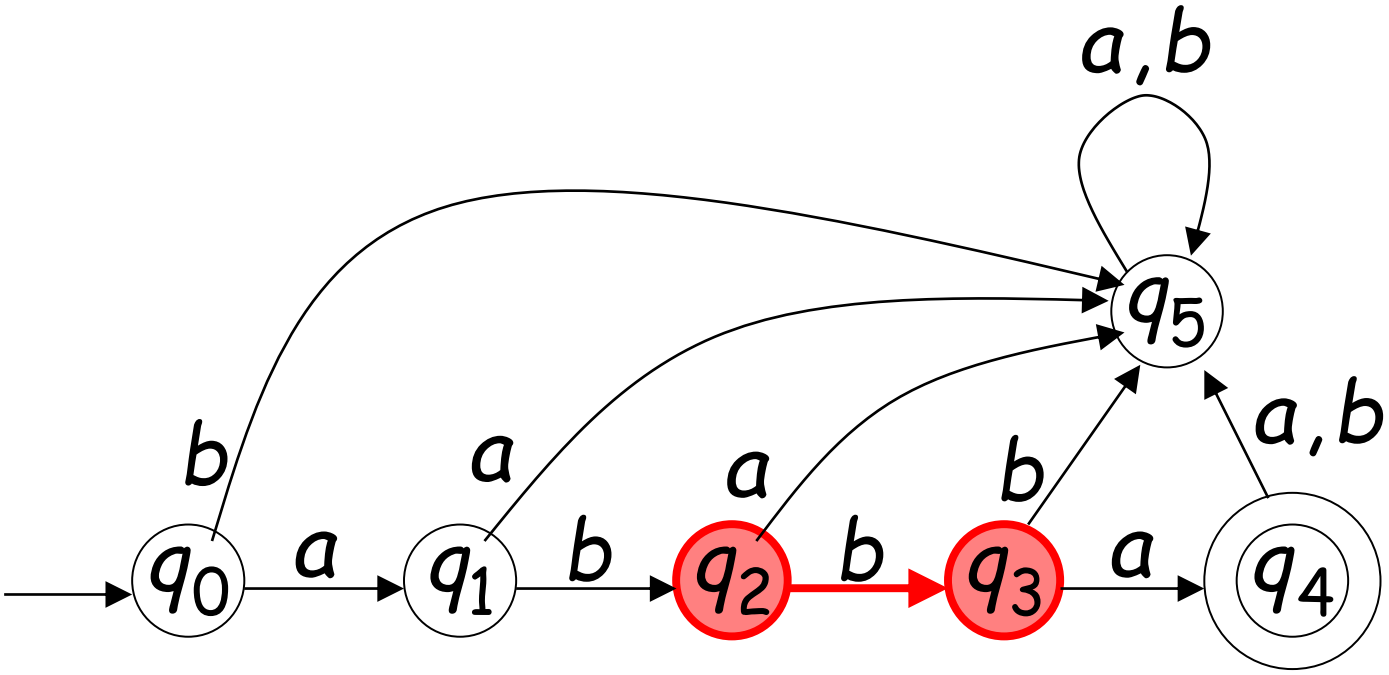
$$\delta(q_0, a) = q_1$$



$$\delta(q_0, b) = q_5$$



$$\delta(q_2, b) = q_3$$

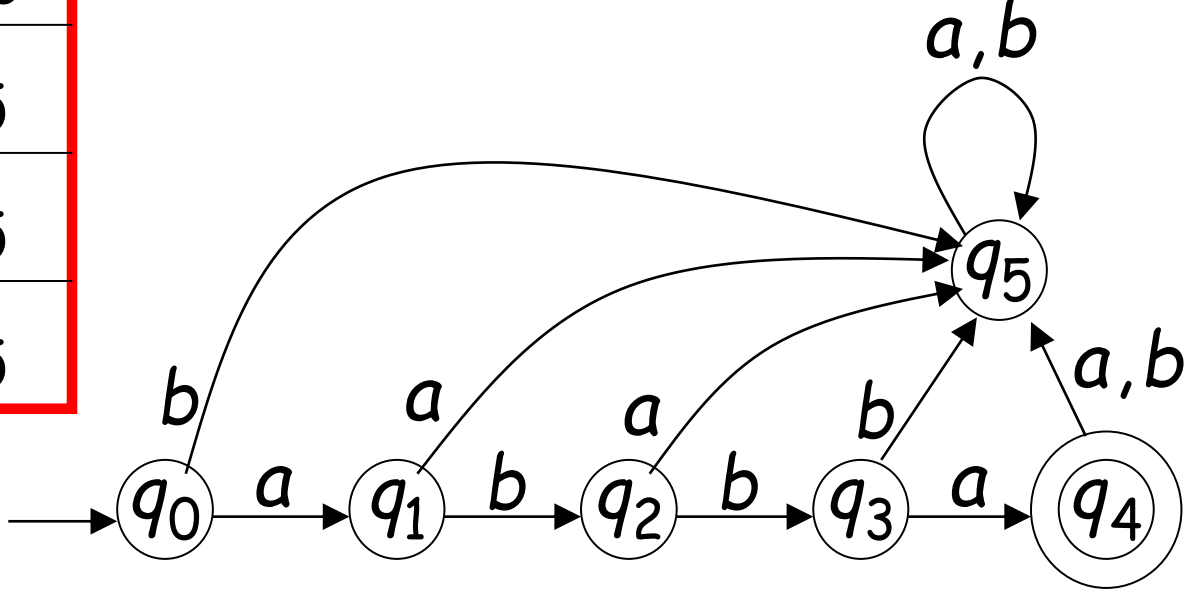


# Transition Table for $\delta$

symbols

states

| $\delta$ | $a$ | $b$ |
|---|---|---|
| $q_0$ | $q_1$ | $q_5$ |
| $q_1$ | $q_5$ | $q_2$ |
| $q_2$ | $q_5$ | $q_3$ |
| $q_3$ | $q_4$ | $q_5$ |
| $q_4$ | $q_5$ | $q_5$ |
| $q_5$ | $q_5$ | $q_5$ |



# Extended Transition Function

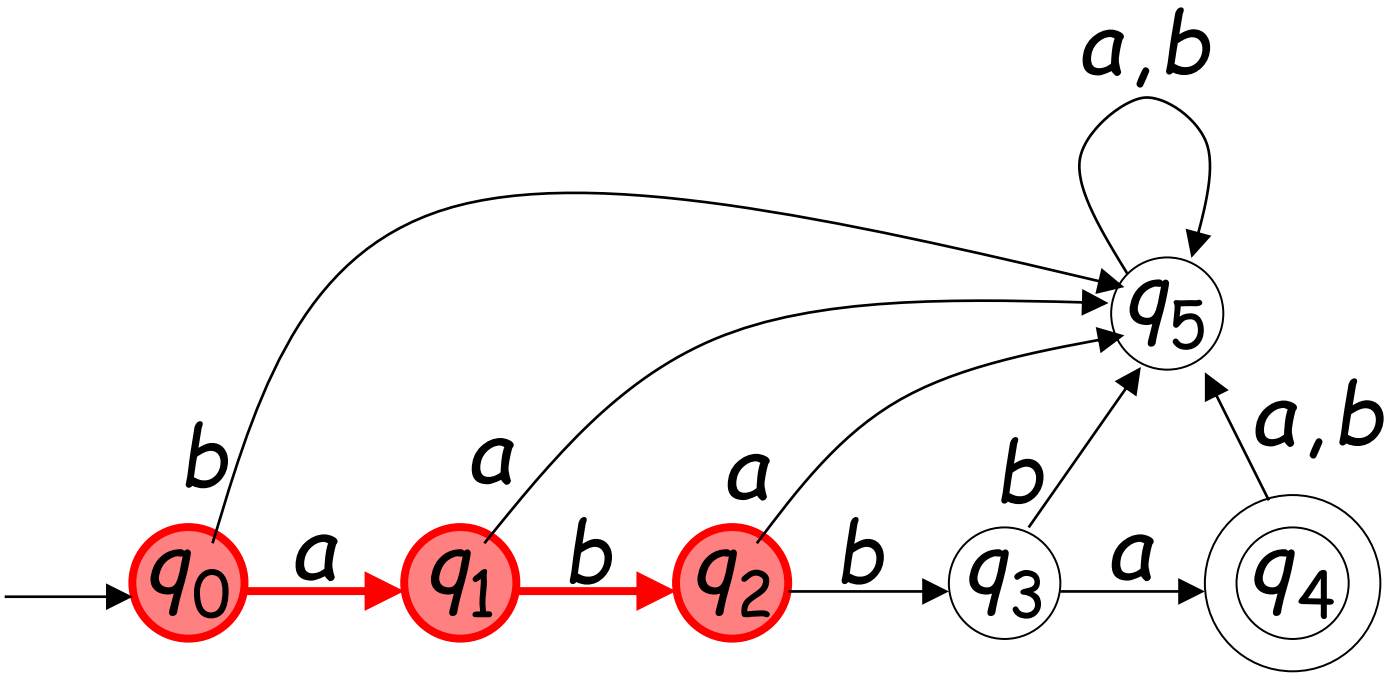
$$\delta^* : Q \times \Sigma^* \rightarrow Q$$

$$\delta^*(q, w) = q'$$

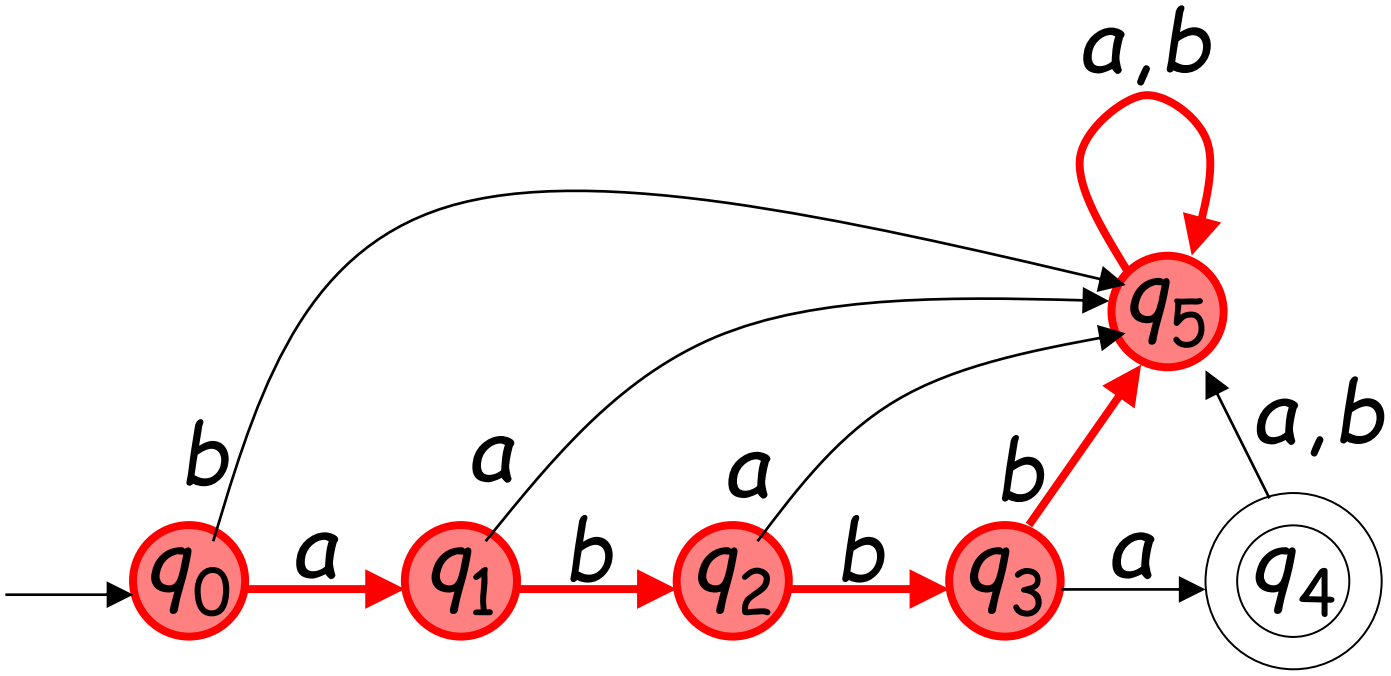
Describes the resulting state

after scanning string $w$ from state $q$

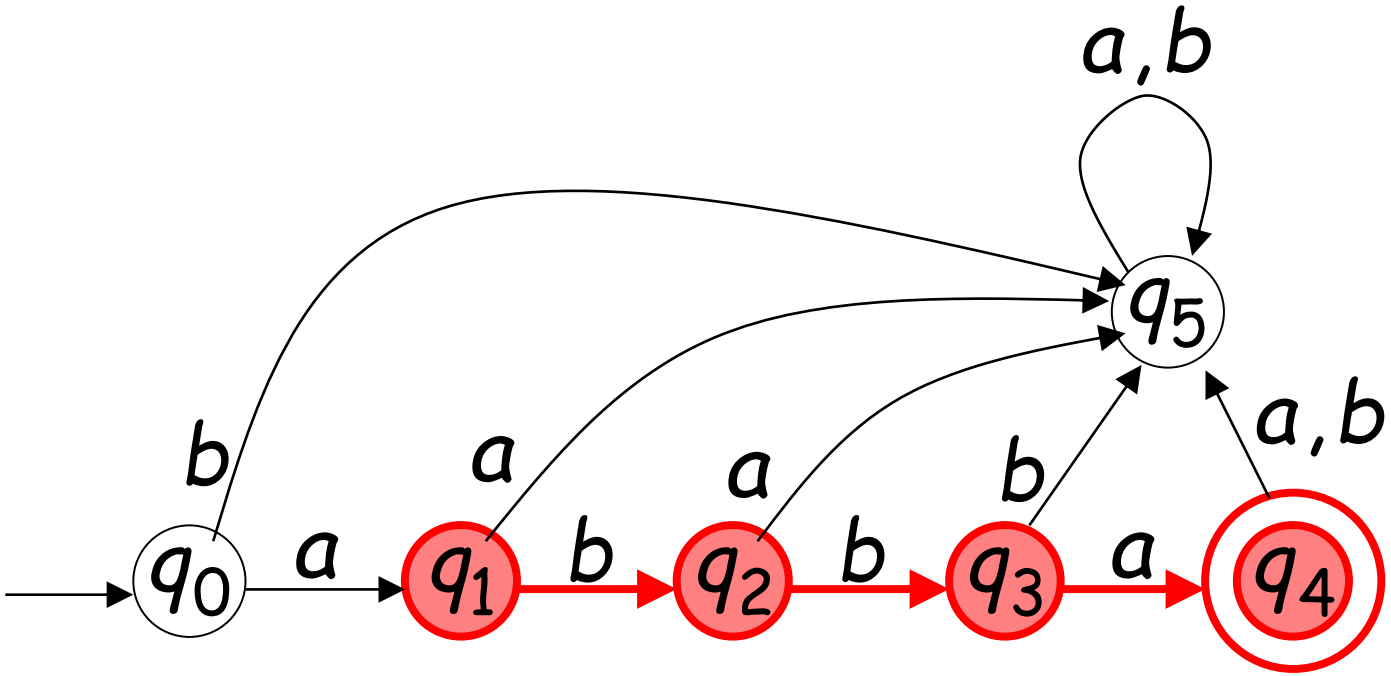
Example: $\delta^*(q_0, ab) = q_2$



$$\delta^*(q_0, abbbaa) = q_5$$



$$\delta^*(q_1, bba) = q_4$$



Special case:

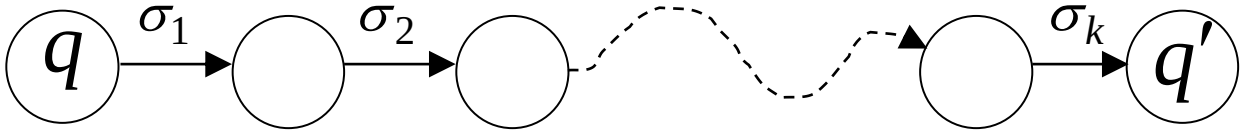
for any state $q$

$$\delta^*(q, \varepsilon) = q$$

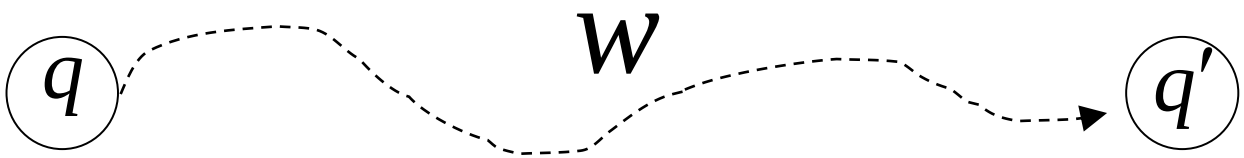
In general: $\delta^*(q, w) = q'$

implies that there is a walk of transitions

$$w = \sigma_1 \sigma_2 \cdots \sigma_k$$




states may be repeated



# Language Recognized by DFA

Language recognized by DFA $M$

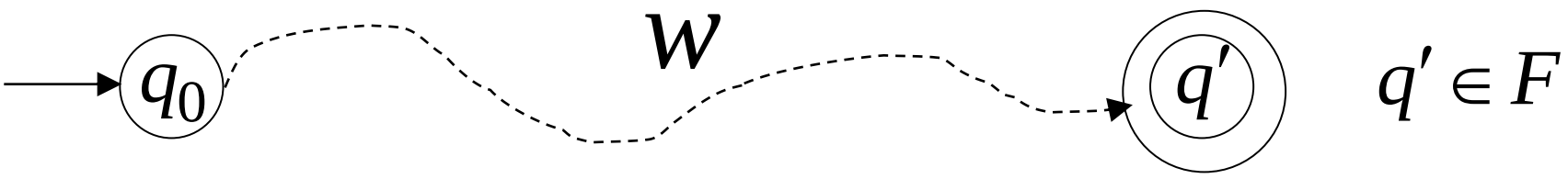
is denoted as $L(M)$ and contains all the strings accepted by $M$

Since, the term accept has different meanings, when we refer to machines accepting strings and machines accepting languages, we prefer the term recognize for languages in order to avoid confusion.

For a DFA $M = (Q, \Sigma, \delta, q_0, F)$

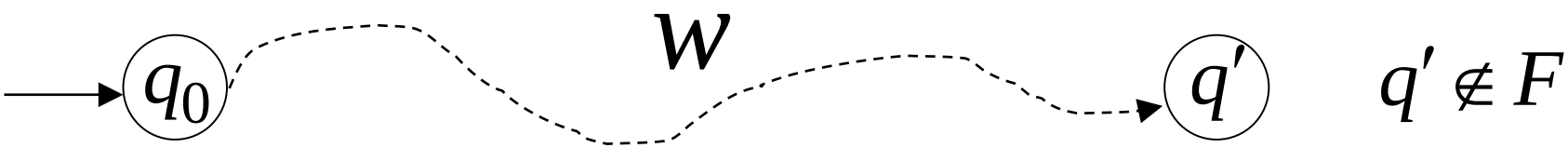
Language recognized by $M$ :

$$L(M) = \{w \in \Sigma^* : \delta^*(q_0, w) \in F\}$$



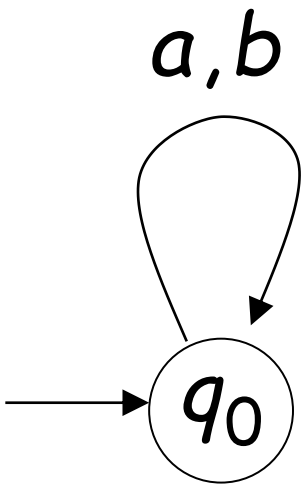
## Language rejected by $M$ :

$$\overline{L(M)} = \left\{ w \in \Sigma^* : \delta^*(q_0, w) \notin F \right\}$$



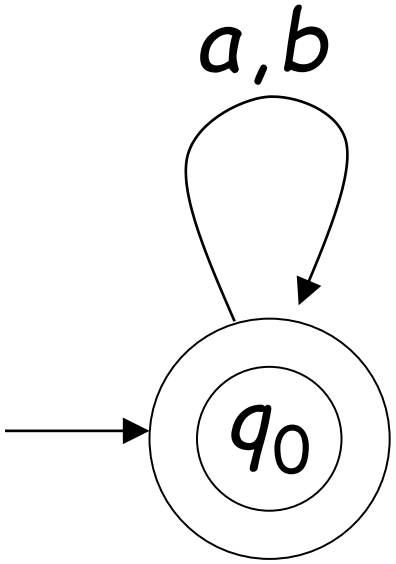
# More DFA Examples

$$\Sigma = \{a, b\}$$




$$L(M) = \{\ \}$$

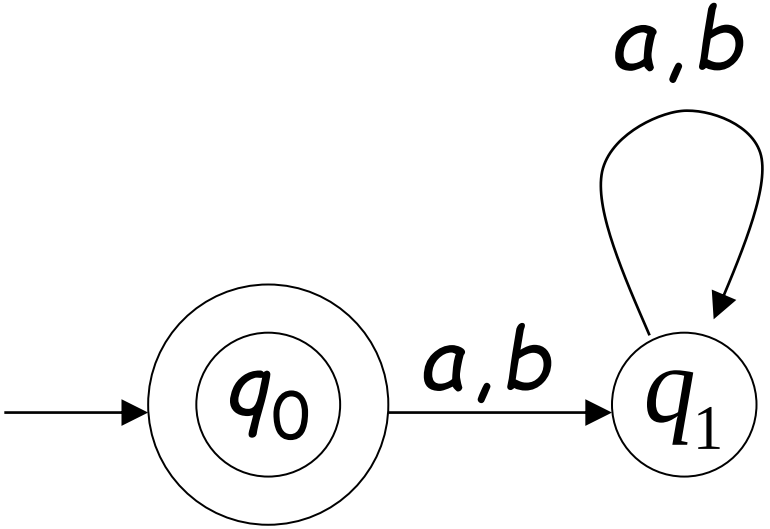
Empty language




$$L(M) = \Sigma^*$$

All strings

$$\Sigma = \{a, b\}$$




$$L(M) = \{\varepsilon\}$$

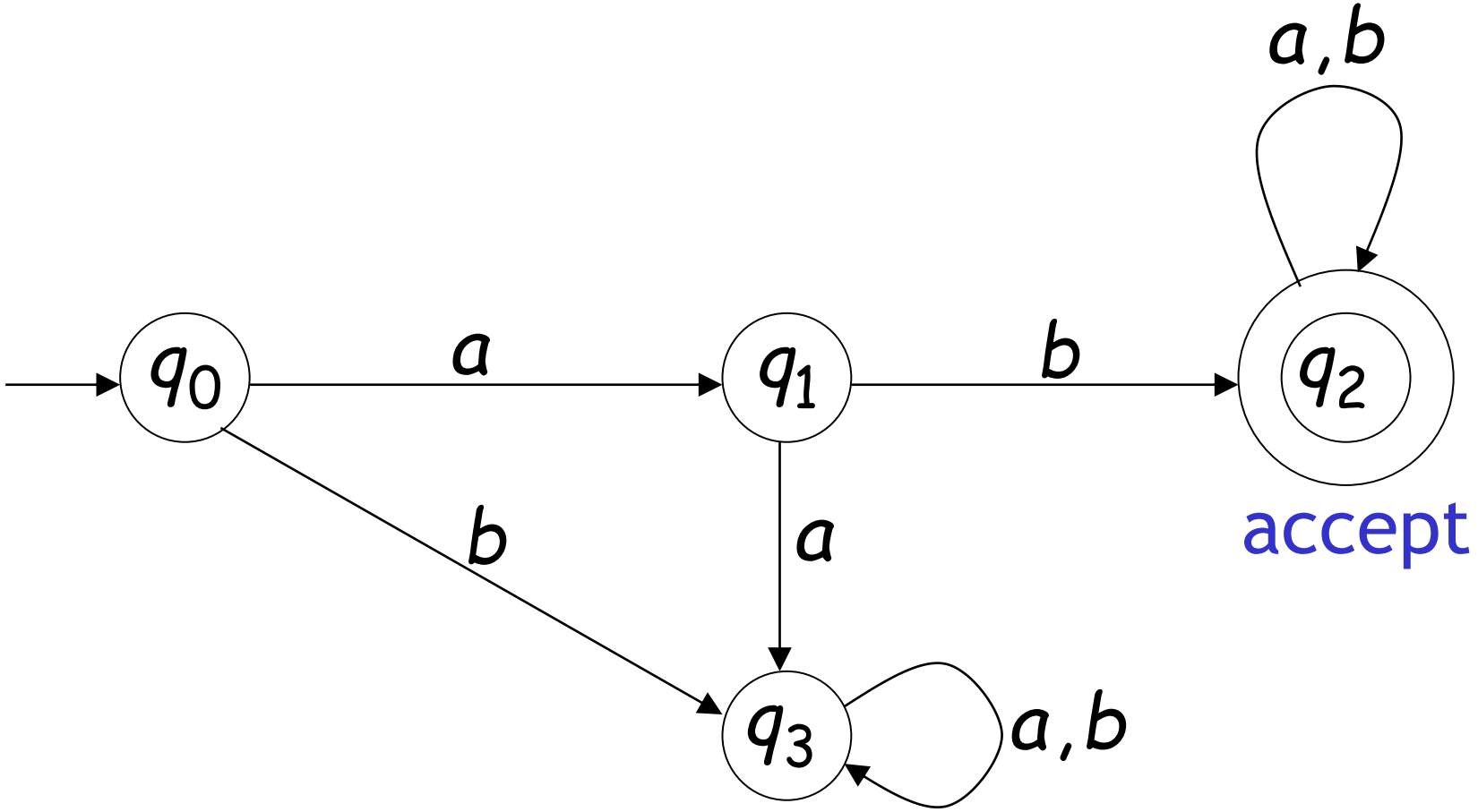
Language of the empty string

$$\Sigma = \{a, b\}$$

$L(M)$= { all strings with prefix $ab$ }

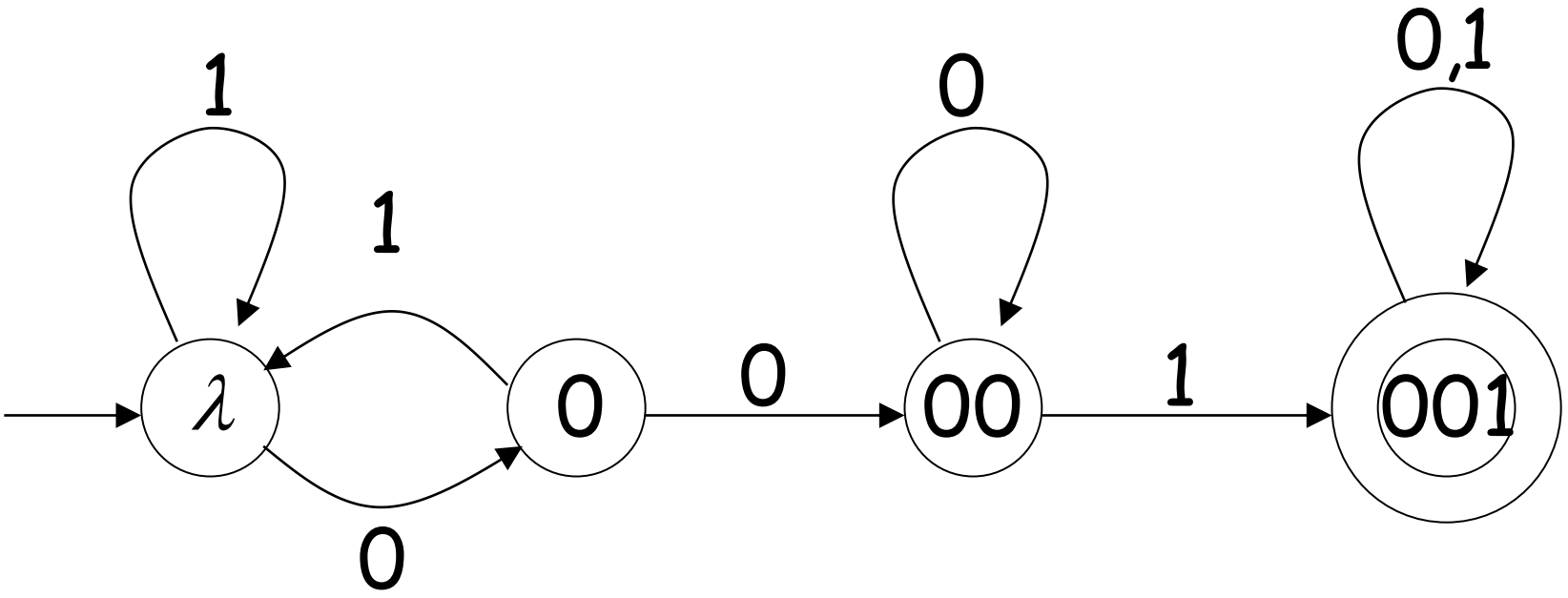
$$\Sigma = \{a, b\}$$

$L(M)$ = { all strings with prefix $ab$ }



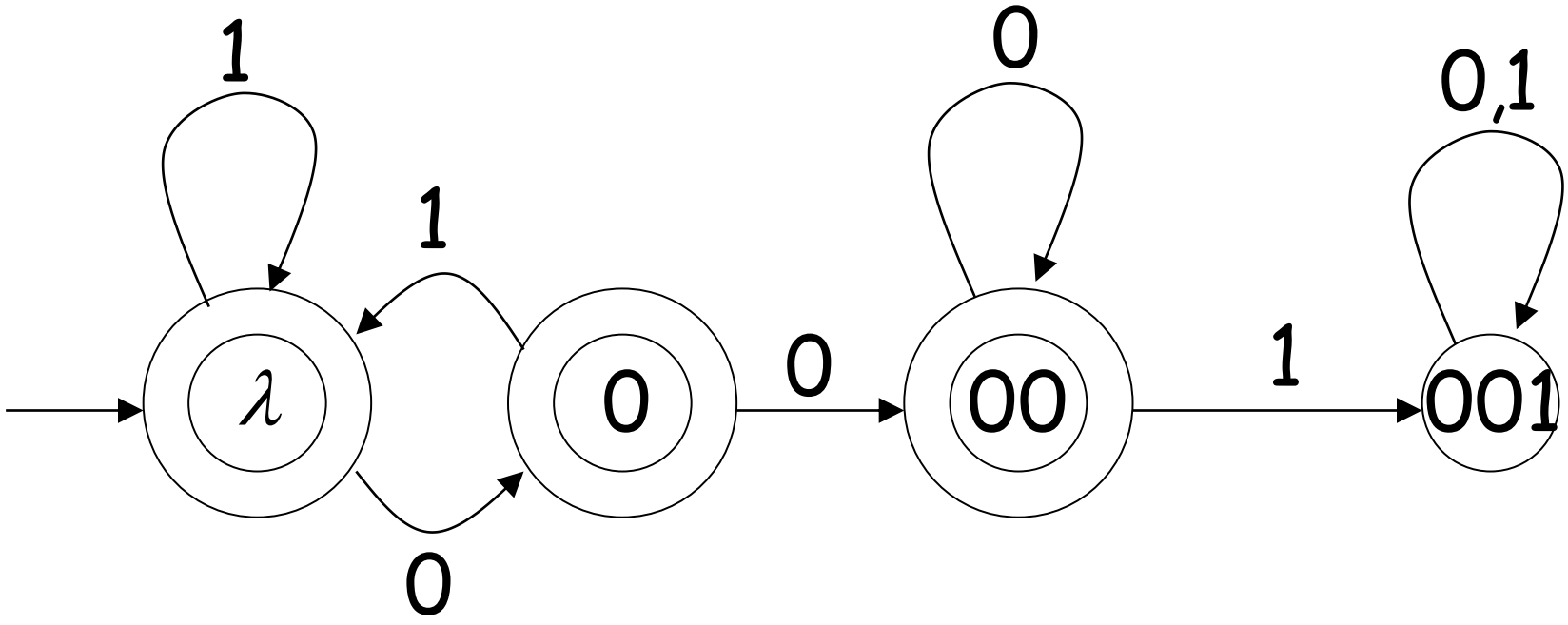
$\Sigma = \{0,1\}$

$L(M)$ = { all binary strings containing substring 001 }

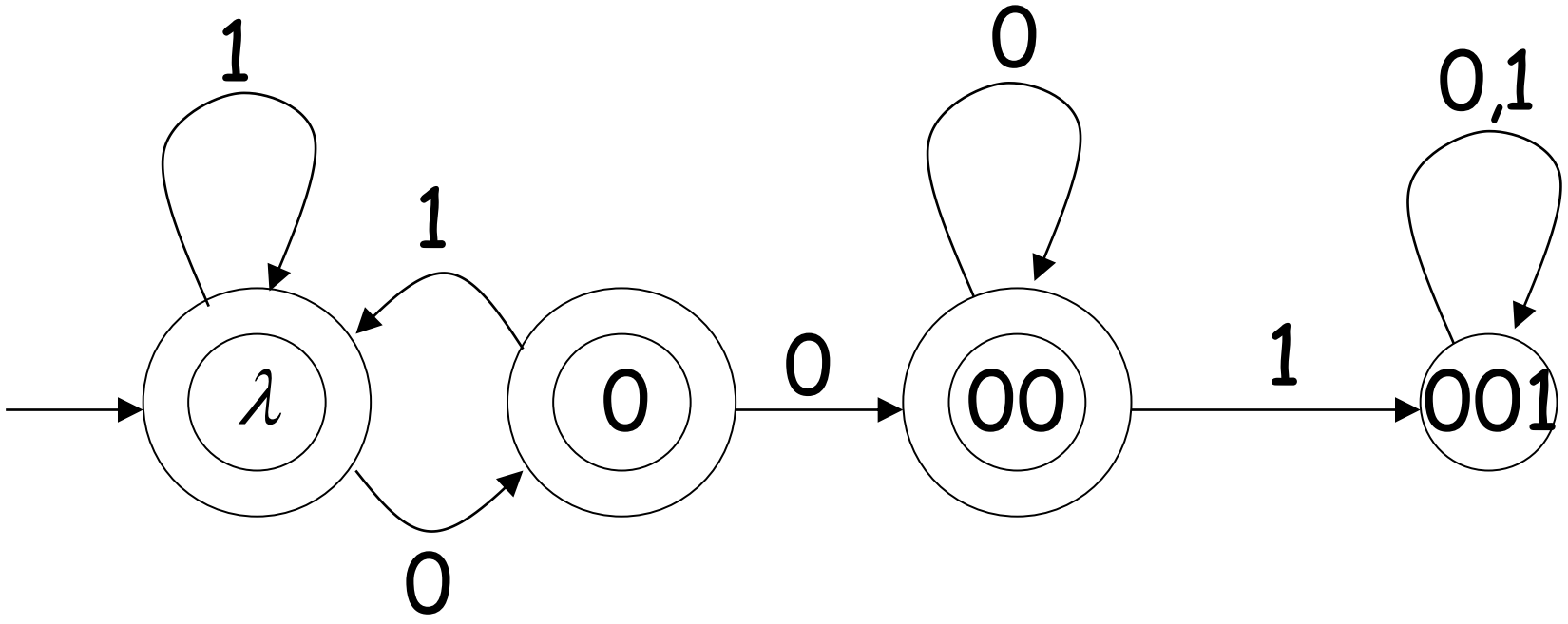


$\Sigma = \{0,1\}$

$L(M)$ = { all binary strings without substring 001 }

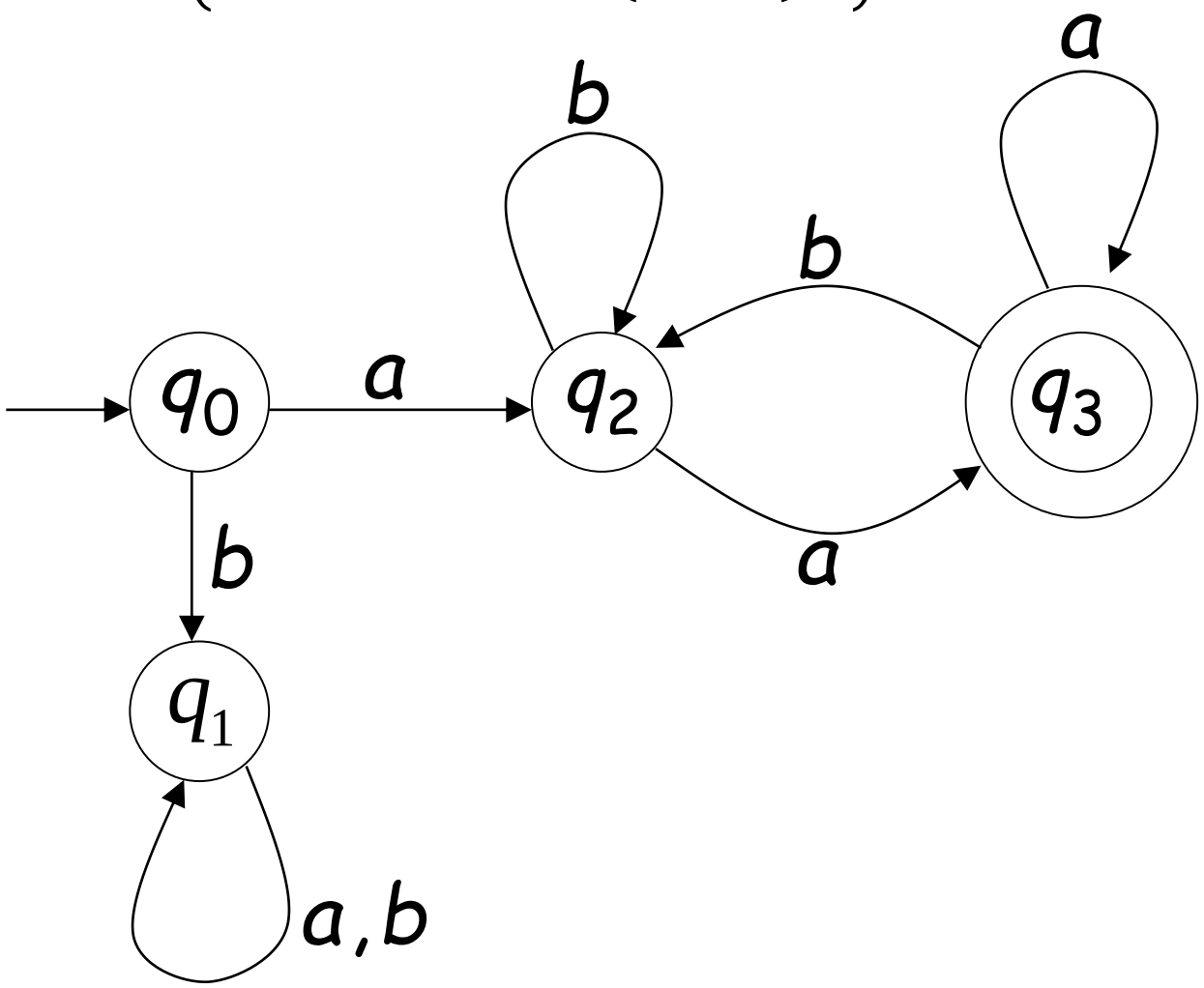


$L(M)$ = { all binary strings without substring 001 }



$$\Sigma = \{a, b\}$$

$$L(M) = \left\{awa : w \in \{a,b\}^*\right\}$$



# Regular Languages

Definition:

A language $L$ is regular if there is a DFA $M$ that recognizes it ($L(M) = L$)

The languages recognized by all DFAs form the family of regular languages

# Example regular languages:

$\{abba\}$ $\{\lambda, ab, abba\}$

$\{a^n b : n \geq 0\}$ $\{awa : w \in \{a,b\}^*\}$

{ all strings in {a,b}* with prefix $ab$ }

{ all binary strings without substring 001 }

$\{x : x \in \{1\}^* \text{ and } x \text{ is even}\}$

$\{\}$ $\{\varepsilon\}$ $\{a,b\}^*$

There exist DFAs that recognizes these languages (see previous slides).

There exist languages which are not Regular:

$$L = \{a^n b^n : n \geq 0\}$$

$$ADDITION = \{x + y = z \ : x = 1^n, y = 1^m, z = 1^k, \ n + m = k\}$$

There are no DFAs that recognize these languages.

(We will prove this later)